Geometric Reduction, Killing-Response Reconstruction, and Spectral Tomography for Navier–Stokes Flow on the Round Three-Sphere

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We develop a unified geometric and spectral theory for incompressible Navier–Stokes flow on the round three-sphere. The common hinge is the Killing–Beltrami interaction on the signed curl decomposition. First, the projected quadratic convection vanishes on every single signed curl shell, so the full Grassmannian of its linear subspaces consists of exact finite-dimensional Navier–Stokes reductions. We study the resulting moduli under the $SO(4)$ isometry action, derive a real-algebraic leakage functional for multi-shell reductions, and obtain its infinitesimal deformation equation. Killing drift transports these reductions by a projector Lax equation; for constant drift we give an exact Grassmannian geodesic criterion, diagonalize the shell generator, derive the finite Diophantine resonance condition, and prove determinant-one shell transport. Second, the six convective responses to Killing generators factor the Stokes operator. After $A^{-1/2}$ normalization they form a Parseval operator-valued frame. Either chiral triplet is uniformly stable, two orthonormal same-chirality probes determine the entire non-Killing velocity, and a single probe is invertible exactly on odd signed-curl levels. Gram compression yields a Casimir response tensor with exact chiral trace splitting. Stokes weighting produces a positive Hankel moment hierarchy, while curl weighting produces a positive matrix-valued spectral measure. Because the signed curl spectrum lies on an integer lattice, its operator-valued characteristic function is periodic; Fourier inversion recovers every signed-shell response tensor, and its scalar trace recovers all signed-shell energies for arbitrary finite-dissipation states. Under linear Stokes evolution this characteristic function satisfies an exact matrix heat equation on the response circle. Combining the reduction and response branches further yields Clebsch–Gordan parity and locality rules for nonlinear shell interactions, one- and two-probe certificates for the full leakage variety, a master response-circle functional calculus, and a Fisher–Rao spectral-selection law whose nonlinear extension separates viscous gradient drift from conservative shell transfer.

I. INTRODUCTION

The round three-sphere is a particularly rigid background for incompressible hydrodynamics: its isometry group is $SO(4)$, its Killing algebra splits into two commuting $\mathfrak{su}(2)$ factors, and the divergence-free spectrum admits a signed curl decomposition. With deformation-tensor viscosity these structures interact directly with the projected Navier–Stokes nonlinearity.

The purpose of this article is to treat two consequences of the same Killing–Beltrami algebra within one theorem chain. The first is a *reduction problem*: which finite-dimensional subspaces are exact Navier–Stokes systems, how are they organized geometrically, and how do Killing drifts move them? The second is an *information problem*: how much of a velocity field can be recovered from its convective responses to the six Killing generators?

The reduction problem begins with a complete cancellation on a single signed curl shell. Since the Stokes operator is scalar there and the polarized projected convection vanishes, every linear subspace of that shell is an exact pressure-linear reduction. Thus exact reductions occur in Grassmannian families rather than as isolated modes. The physical $SO(4)$ action produces quotient moduli, while cross-shell interactions lead to a real-algebraic zero-leakage locus on products of Grassmannians. Killing drift gives a natural nonabelian transport on these moduli. Its projector satisfies a Lax equation, but a constant drift orbit is geodesic only under an additional block-compatibility condition. The same representation theory gives the exact fixed-drift spectrum and its resonance arithmetic.

The information problem begins from the quadratic Casimir factorization of the Stokes operator by six Killing-response operators. After Stokes normalization these responses form a Parseval operator-valued frame. The redundancy is explicit: either chiral triplet remains a stable frame, any two orthonormal probes in one chiral factor are globally injective, and the one-probe problem has an exact odd/even shell parity law.

Compressing the response fields to their 6×6 Gram operator yields a Casimir response geometry. Applying powers of the Stokes operator produces a positive moment hierarchy whose traces are the Stokes spectral moments. Applying powers of curl instead retains chirality and yields a positive matrix-valued measure on the signed curl spectrum. Since $\mu_{k,\sigma} = \sigma(k+2)/R$, the associated unitary curl group is periodic with period $2\pi R$; consequently a single operator-valued function on this response circle admits exact Fourier inversion.

The individual ingredients have established relatives: invariant helical subspaces and special exact solutions occur in Navier–Stokes theory [6, 19]; Killing fields in deformation-tensor Navier–Stokes equations were studied by Samavaki and Tuomela [21]; Grassmann geome-

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try, operator-valued frames, Lie-group Riesz transforms, and matrix-valued moment problems have independent abstract theories [4, 5, 16, 17, 22]. The contribution here is the exact realization of these structures by one $SO(4)$-equivariant Navier–Stokes system and the explicit theorems linking reduction geometry, minimal observability, and spectral tomography. No historical priority is inferred from the literature search; the claims are the proved statements below.

Relation to the three public Zenodo preprints

The present manuscript belongs to a three-stage public preprint program.

The earliest work, Ref. [3], was publicly deposited on Zenodo on September 20, 2026. It introduced the general finite-dimensional closure criterion and the zero-defect viewpoint, developed the maximal-torus, Killing, and signed-Beltrami sectors, established spectral-escape/projector bounds, and kept the forced singular-transfer construction explicitly conditional.

The second work, Ref. [2], was publicly deposited on Zenodo on September 21, 2026. It isolated the exact Killing–Beltrami intertwining law, shell-diagonal Killing precession, the associated connection and curvature, Killing-saturated rigidity, the quadratic Casimir factorization, a finite toric classification, and sharp homogeneous-projector norms. That paper is not superseded by the unified manuscript: several of these classification and homogeneous-bundle results are not reproduced below.

The third work, Ref. [1], was also publicly deposited on September 21, 2026 and is the immediate Version 1.0 baseline for the unified reduction/response theory. It developed Grassmannian reduction moduli, multi-shell leakage geometry, Killing-precession geometry, lossless Killing-response reconstruction, Casimir response moments, and curl-Fourier tomography.

The additional results collected in Sec. IX were derived subsequently. In particular, the Clebsch–Gordan interaction selection rule there answers a limitation explicitly left open in Ref. [2]: that paper did not infer a chiral shell selection rule without an explicit Clebsch–Gordan analysis. The new section should therefore be read as a subsequent extension of the three public preprints, not as a retrospective claim about their original contents.

II. GEOMETRIC AND SPECTRAL SETTING

Let (S^3_R, g) be the round three-sphere of radius R, with

$$V_R = \mathrm{Vol}(S^3_R) = 2\pi^2 R^3.$$

We use the deformation-tensor viscosity

$$\begin{aligned} \partial_t u + \nabla_u u - \nu \mathcal{L}_g u + \mathrm{grad}_g\, p &= f, \\ \mathrm{div}_g\, u &= 0, \\ \mathcal{L}_g u &= 2\ \mathrm{div}_g\, \mathrm{Def}_g\, u. \end{aligned}$$

Let $\mathbb{P}$ denote the Leray projection and put

$$\begin{aligned} A &= -\mathbb{P}\mathcal{L}_g, \\ B(u,v) &= \mathbb{P}\nabla_u v, \\ Q(u,v) &= \frac{1}{2}\{B(u,v) + B(v,u)\}. \end{aligned}$$

Then

$$\partial_t u + \nu A u + Q(u,u) = \mathbb{P} f. \tag{1}$$

The divergence-free L^2 space decomposes as

$$L^2_\sigma(TS^3_R) = \widehat{\bigoplus}_{k\geq 0,\ \sigma=\pm} E_{k,\sigma}, \tag{2}$$

where

$$\mathrm{curl}\, u = \mu_{k,\sigma} u, \qquad \mu_{k,\sigma} = \sigma \frac{k+2}{R}, \tag{3}$$

$$A u = \kappa_k u, \qquad \kappa_k = \frac{k(k+4)}{R^2}, \tag{4}$$

$$\dim E_{k,\sigma} = d_k = (k+1)(k+3). \tag{5}$$

These formulas are consistent with the standard spectrum of differential forms on spheres [15] and with the signed-shell conventions of Ref. [3].

We first record the cancellation that drives everything that follows.

Lemma II.1 (Polarized single-shell cancellation)**.** *For all $u, v \in E_{k,\sigma}$,*

$$Q(u,v) = 0. \tag{6}$$

Proof. On an oriented Riemannian three-manifold,

$$\nabla_u v + \nabla_v u = \mathrm{grad}_g\langle u, v\rangle - u \times \mathrm{curl}\, v - v \times \mathrm{curl}\, u.$$

If u and v belong to the same signed curl shell, then $\mathrm{curl}\, u = \mu u$ and $\mathrm{curl}\, v = \mu v$ with the same scalar μ. Hence

$$-u \times \mathrm{curl}\, v - v \times \mathrm{curl}\, u = -\mu u \times v - \mu v \times u = 0.$$

Thus $\nabla_u v + \nabla_v u$ is a gradient, and its Leray projection vanishes. □

Remark II.2. For $u = v$, Theorem II.1 is the usual Beltrami observation that $\nabla_u u$ is a gradient. The polarized statement is stronger: it removes the interaction between *any two* vectors in the same signed shell.

III. EXACT-REDUCTION MODULI AND LEAKAGE GEOMETRY

A. The Grassmannian exact-reduction theorem

Fix (k, σ) and abbreviate

$$E = E_{k,\sigma}, \qquad D = \dim E = d_k.$$

All Grassmannians below are real.

Theorem III.1 (Grassmannian family of exact reductions)**.** *Let $1 \le d \le D$. Every point*

$$L \in \mathrm{Gr}(d, E)$$

is an exact d-dimensional reducing space for (1):

$$AL \subset L, \qquad Q(L, L) = 0. \tag{7}$$

For unforced dynamics and $u_0 \in L$,

$$u(t) = e^{-\nu\kappa_k t} u_0. \tag{8}$$

If $\mathbb{P}f(t) \in L$ for all t, then

$$u(t) = e^{-\nu\kappa_k t} u_0 + \int_0^t e^{-\nu\kappa_k (t-s)} \mathbb{P}f(s)\, \mathrm{d}s. \tag{9}$$

Proof. Equation (4) gives $A|_E = \kappa_k \mathrm{Id}_E$, hence $AL \subset L$ for every subspace $L \subset E$. Equation (6) gives $Q(L, L) = 0$. Restricting (1) to L therefore leaves the linear scalar equation

$$\partial_t u + \nu\kappa_k u = \mathbb{P}f.$$

The formulas follow by variation of constants. □

Corollary III.2 (No finite low-dimensional atlas)**.** *For every $d \ge 1$ there are infinitely many exact d-dimensional linear Navier–Stokes reductions on S^3_R. For every $1 \le d \le 8$ there are continuous families already inside a single signed curl shell.*

Proof. Choose k so that $d_k > d$. Then $\mathrm{Gr}(d, E_{k,\sigma})$ is a positive-dimensional compact manifold of dimension $d(d_k - d)$. □

Remark III.3. The theorem concerns exact *linear* invariant subspaces. It is different from the local nonlinear spectral submanifolds used in modern model reduction [7]. Here invariance is global on the subspace and no perturbative graph construction is required.

B. Tautological Navier–Stokes dynamics

Let

$$\mathcal{G}_{D,d} := \mathrm{Gr}(d, E)$$

and let

$$\pi : \mathscr{S}_{D,d} \longrightarrow \mathcal{G}_{D,d}$$

be the tautological bundle,

$$\mathscr{S}_{D,d} = \{(L, u) \in \mathcal{G}_{D,d} \times E : \ u \in L\}.$$

Theorem III.4 (Universal tautological semigroup)**.** *For each $t \ge 0$, define*

$$\Phi_t : \mathscr{S}_{D,d} \longrightarrow \mathscr{S}_{D,d}, \qquad \Phi_t(L, u) = \big(L, e^{-\nu\kappa_k t} u\big). \tag{10}$$

Then $(\Phi_t)_{t \ge 0}$ is a smooth bundle semigroup over the identity on $\mathcal{G}_{D,d}$, and its restriction to every fiber is exactly the unforced Navier–Stokes flow on that reduction.

The construction is $SO(4)$-equivariant.

Proof. Fiber preservation and equality with the Navier–Stokes flow are Theorem III.1. The formula is smooth and satisfies $\Phi_{t+s} = \Phi_t \circ \Phi_s$. Since the $SO(4)$ action preserves the round metric, curl shells, and the Stokes operator,

$$g \cdot \Phi_t(L, u) = (gL, e^{-\nu\kappa_k t} gu) = \Phi_t(gL, gu).$$

□

Remark III.5. The base point L labels the exact reduced model; the fiber variable is the fluid state within that model. Thus the tautological bundle packages all d-dimensional same-shell exact reductions into one geometric dynamical object.

C. Isometry moduli and orbit dimension

The isometry group $G = SO(4)$ acts orthogonally on $E_{k,\sigma}$ and hence smoothly on $\mathrm{Gr}(d, E_{k,\sigma})$.

Definition III.6. The moduli space of d-dimensional same-shell reductions up to isometry is

$$\mathscr{M}_{k,\sigma,d} := \mathrm{Gr}(d, E_{k,\sigma})/SO(4). \tag{11}$$

Theorem III.7 (Compact quotient and dimension lower bound)**.** *The quotient $\mathscr{M}_{k,\sigma,d}$ is compact and Hausdorff. Its principal orbit stratum has dimension*

$$d(d_k - d) - r_{\mathrm{pr}}, \tag{12}$$

where $r_{\mathrm{pr}} \le 6$ is the dimension of a principal $SO(4)$ orbit. Consequently,

$$\boxed{\dim \mathscr{M}_{k,\sigma,d} \ge d(d_k - d) - 6.} \tag{13}$$

In particular, if $d(d_k - d) > 6$, there is a positive-dimensional continuum of pairwise isometrically inequivalent exact reductions.

Proof. The real Grassmannian is compact and $SO(4)$ is compact, so the orbit space is compact Hausdorff. By the principal orbit type theorem, a dense open subset of the Grassmannian consists of orbits of one principal type. If their dimension is r_{pr}, then the quotient of this principal stratum has dimension

$$\dim \mathrm{Gr}(d, D) - r_{\mathrm{pr}} = d(D - d) - r_{\mathrm{pr}}.$$

Since $\dim SO(4) = 6$, one has $r_{\mathrm{pr}} \le 6$. □

Corollary III.8 (Positive-dimensional moduli in the first non-Killing shell)**.** *For $k = 1$, one has $d_1 = 8$. Hence for every proper dimension $1 \le d \le 7$,*

$$\dim \mathscr{M}_{1,\sigma,d} \ge d(8-d) - 6 > 0.$$

Thus even the first non-Killing signed curl shell contains continuous isometry moduli of exact reductions in every proper dimension.

Remark III.9. The quotient need not be a smooth manifold globally because stabilizer type can jump. The theorem uses only the principal stratum, so no global freeness assumption is made.

D. A concentration landscape on the Grassmannian

Let L be any d-dimensional subspace of smooth vector fields on S^3_R, and let Π_L be its L^2-orthogonal projector. Choose an orthonormal basis $\{\phi_j\}_{j=1}^d$ of L. The projector kernel is

$$K_L(x,y) = \sum_{j=1}^{d} \phi_j(x) \otimes \phi_j(y)^\flat. \tag{14}$$

At the diagonal $K_L(x,x)$ is a positive semidefinite endomorphism of the three-dimensional tangent space.

Lemma III.10 (Evaluation formula)**.** *For the evaluation map $E_x : L \to T_x S^3_R$, one has*

$$K_L(x,x) = E_x E_x^*,$$

and therefore

$$\|\Pi_L\|^2_{2\to\infty} = \sup_{x \in S^3_R} \lambda_{\max}\big(K_L(x,x)\big). \tag{15}$$

Moreover,

$$\int_{S^3_R} \operatorname{tr} K_L(x,x)\, \mathrm{d}V = d. \tag{16}$$

Proof. The kernel identity follows directly from an orthonormal basis. The $L^2 \to L^\infty$ norm is the supremum of the evaluation norms, and $\|E_x\|^2 = \lambda_{\max}(E_x E_x^*)$. Finally,

$$\int \operatorname{tr} K_L(x,x)\, \mathrm{d}V = \sum_{j=1}^{d} \|\phi_j\|_2^2 = d.$$

□

Theorem III.11 (Universal concentration floor)**.** *For every d-dimensional subspace L of smooth vector fields on S^3_R,*

$$\boxed{\|\Pi_L\|_{2\to\infty} \ge \sqrt{\frac{d}{3V_R}},} \tag{17}$$

$$\boxed{\|\Pi_L\|_{1\to\infty} \ge \frac{d}{3V_R}.} \tag{18}$$

In particular these inequalities hold for every point of every exact Grassmannian family $\mathrm{Gr}(d, E_{k,\sigma})$.

Proof. By (16), there exists x with

$$\operatorname{tr} K_L(x,x) \ge \frac{d}{V_R}.$$

Since $K_L(x,x)$ acts on a three-dimensional space,

$$\lambda_{\max}\big(K_L(x,x)\big) \ge \frac{1}{3} \operatorname{tr} K_L(x,x) \ge \frac{d}{3V_R}.$$

Equation (17) follows from (15).

For the $L^1 \to L^\infty$ norm, the continuous finite-rank kernel satisfies

$$\|\Pi_L\|_{1\to\infty} = \sup_{x,y} \|K_L(x,y)\|_{\mathrm{op}}.$$

The diagonal is included in the supremum, so the same point x gives (18). □

Definition III.12. For $L \in \mathrm{Gr}(d, E_{k,\sigma})$, define the dimensionless concentration defects

$$\mathcal{C}_2(L) := \frac{3V_R}{d}\|\Pi_L\|^2_{2\to\infty}, \qquad \mathcal{C}_1(L) := \frac{3V_R}{d}\|\Pi_L\|_{1\to\infty}. \tag{19}$$

Then $\mathcal{C}_1, \mathcal{C}_2 \ge 1$.

These functions are invariant under the physical $SO(4)$ action and therefore descend to the isometry moduli space (11). They provide quantitative functions on the moduli of exact reductions.

E. Random Grassmannian reductions

We now equip $\mathrm{Gr}(d, E)$ with its normalized $O(E)$-invariant probability measure. This ensemble is not a physical symmetry average; it is the canonical uniform measure on the parameter space of exact reductions.

Let $P_L : E \to E$ denote the orthogonal projector onto L.

Lemma III.13 (First moment of a random subspace projector)**.** *If L is uniformly distributed in $\mathrm{Gr}(d, E)$ and $D = \dim E$, then*

$$\mathbb{E}\, P_L = \frac{d}{D} \mathrm{Id}_E. \tag{20}$$

Proof. The expectation commutes with every orthogonal transformation of E, so it is a scalar multiple of the identity. Taking the trace gives the scalar d/D. □

Let $K_E(x,y)$ be the orthogonal projector kernel of the full shell. By $SO(4)$ homogeneity and irreducibility of the tangent isotropy,

$$K_E(x,x) = \frac{D}{3V_R} \mathrm{Id}_{T_x S^3_R}. \tag{21}$$

Theorem III.14 (Ensemble isotropy). *For a uniformly random $L \in \mathrm{Gr}(d, E_{k,\sigma})$,*

$$\mathbb{E}\, K_L(x,y) = \frac{d}{D} K_E(x,y), \tag{22}$$

$$\boxed{\mathbb{E}\, K_L(x,x) = \frac{d}{3V_R}\mathrm{Id}_{T_x S^3_R}.} \tag{23}$$

Thus the ensemble of exact reductions is pointwise isotropic in expectation, even though an individual reduction need not be homogeneous.

Proof. Write $\mathrm{ev}_x : E \to T_x S^3_R$ for evaluation. Then

$$K_L(x,y) = \mathrm{ev}_x\, P_L\, \mathrm{ev}_y^*.$$

Take expectation and use (20); the diagonal formula follows from (21). □

The second moment is also explicit.

Theorem III.15 (Exact pointwise projector variance). *Let $D = d_k$ and $1 \le d \le D$. For every $x \in S^3_R$,*

$$\boxed{\mathbb{E}\left\| K_L(x,x) - \frac{d}{3V_R}\mathrm{Id}\right\|^2_{\mathrm{HS}} = \frac{2d(D-d)(2D-1)}{3V_R^2(D-1)(D+2)}.} \tag{24}$$

The right-hand side is independent of x.

Proof. Fix an orthonormal basis of E and write $P = P_L$. Orthogonal invariance implies

$$\mathbb{E}(P_{ij}P_{kl}) = a\,\delta_{ij}\delta_{kl} + b\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\big). \tag{25}$$

The identities $\mathrm{tr}\, P = d$ and $\mathrm{tr}\, P^2 = d$ give

$$a = \frac{d\{d(D+1)-2\}}{D(D-1)(D+2)}, \tag{26}$$

$$b = \frac{d(D-d)}{D(D-1)(D+2)}. \tag{27}$$

Let $T = \mathrm{ev}_x : E \to T_x S^3_R$ and $H = T^*T$. Then $K_L(x,x) = TPT^*$ and, by (21),

$$\mathrm{tr}\, H = \frac{D}{V_R}, \qquad \mathrm{tr}\, H^2 = \frac{D^2}{3V_R^2}. \tag{28}$$

Contracting (25) yields

$$\mathbb{E}\ \mathrm{tr}(TPT^*)^2 = (a+b)\,\mathrm{tr}\, H^2 + b(\mathrm{tr}\, H)^2. \tag{29}$$

Subtract

$$\left\|\frac{d}{3V_R}\mathrm{Id}\right\|^2_{\mathrm{HS}} = \frac{d^2}{3V_R^2},$$

insert (26)–(28), and simplify. This gives (24). □

Remark III.16. For a fixed proportion d/D, the absolute variance grows only linearly in D, whereas the squared size of the mean diagonal kernel grows quadratically. Thus high-dimensional random exact reductions become relatively isotropic at a fixed point.

F. Multi-shell leakage as a real-algebraic moduli problem

The single-shell Grassmannian is maximally degenerate: every point is exact. Cross-shell interaction creates the genuine classification problem.

Let

$$\mathcal{A} = \{\alpha_1, \dots, \alpha_m\}, \qquad E_{\alpha_j} = E_{k_j,\sigma_j},$$

be finitely many signed shells and set

$$V_{\mathcal{A}} = \bigoplus_{\alpha\in\mathcal{A}} E_\alpha.$$

Fix a dimension vector $\mathbf{d} = (d_\alpha)_{\alpha\in\mathcal{A}}$ with $0 \le d_\alpha \le \dim E_\alpha$, and define

$$\mathcal{G}_{\mathcal{A},\mathbf{d}} := \prod_{\alpha\in\mathcal{A}} \mathrm{Gr}(d_\alpha, E_\alpha). \tag{30}$$

A point $\mathbf{L} = (L_\alpha)_\alpha$ determines

$$L(\mathbf{L}) = \bigoplus_{\alpha\in\mathcal{A}} L_\alpha$$

and its orthogonal projector

$$P_{\mathbf{L}} = \bigoplus_{\alpha\in\mathcal{A}} P_{L_\alpha},$$

extended by zero on $V_{\mathcal{A}}^\perp$.

Because A is scalar on each shell,

$$AL(\mathbf{L}) \subset L(\mathbf{L})$$

for every point of (30). Only quadratic leakage remains.

Definition III.17 (Grassmannian leakage functional). Choose any orthonormal basis $\{e_i\}_{i=1}^N$ of $V_{\mathcal{A}}$. Define

$$\Lambda_{\mathcal{A},\mathbf{d}}(\mathbf{L}) := \sum_{i,j=1}^{N} \|(I - P_{\mathbf{L}})Q(P_{\mathbf{L}}e_i, P_{\mathbf{L}}e_j)\|_2^2. \tag{31}$$

This is the squared Hilbert–Schmidt norm of the bilinear map

$$(I - P_{\mathbf{L}})Q\big|_{L(\mathbf{L})\times L(\mathbf{L})},$$

so it is independent of the chosen orthonormal basis.

Theorem III.18 (Leakage variety). *The function $\Lambda_{\mathcal{A},\mathbf{d}}$ has the following properties.*

(i) It is nonnegative, continuous, real analytic, and $SO(4)$-invariant.

(ii) In the orthogonal-projector realization

$$\mathrm{Gr}(d,E) \simeq \{P \in \mathrm{End}(E) : P^2 = P,\ P^* = P,\ \mathrm{tr}\, P = d\},$$

it is the restriction of a polynomial function of the projector entries.

(iii) Its zero set

$$\mathcal{Z}_{\mathcal{A},\mathbf{d}} := \{\mathbf{L} \in \mathcal{G}_{\mathcal{A},\mathbf{d}} : \Lambda_{\mathcal{A},\mathbf{d}}(\mathbf{L}) = 0\} \tag{32}$$

is exactly the set of exact reducing spaces with the prescribed spectral dimension vector:

$$\mathbf{L} \in \mathcal{Z}_{\mathcal{A},\mathbf{d}} \quad \Longleftrightarrow \quad Q(L(\mathbf{L}), L(\mathbf{L})) \subset L(\mathbf{L}). \tag{33}$$

(iv) $\mathcal{Z}_{\mathcal{A},\mathbf{d}}$ is a compact $SO(4)$-invariant real-algebraic subset of the compact product Grassmannian.

Proof. Nonnegativity and continuity are immediate. The Hilbert–Schmidt expression is independent of basis. Equivariance of the Navier–Stokes bilinear map under isometries gives $SO(4)$ invariance.

For algebraicity, use orthogonal projector coordinates. The vectors $P_{\mathbf{L}} e_i$ depend linearly on the entries of $P_{\mathbf{L}}$, the bilinear map Q is fixed, and the final application of $I - P_{\mathbf{L}}$ is again linear in the projector. Since $Q(V_{\mathcal{A}}, V_{\mathcal{A}})$ spans a finite-dimensional space, the squared L^2 norm in (31) is a finite polynomial expression in those entries.

The sum of squares vanishes exactly when every leakage vector vanishes, which is precisely

$$Q(L(\mathbf{L}), L(\mathbf{L})) \subset L(\mathbf{L}).$$

Stokes invariance is automatic from the shell decomposition, proving (33). Compactness follows because the zero set is closed inside a compact product of Grassmannians. □

Corollary III.19 (Single-shell component)**.** *If $\mathcal{A} = \{\alpha\}$ consists of one signed shell, then*

$$\Lambda_{\mathcal{A},d} \equiv 0, \qquad \mathcal{Z}_{\mathcal{A},d} = \mathrm{Gr}(d, E_\alpha).$$

Corollary III.20 (Complete-shell leakage is a vertex specialization)**.** *Suppose each d_α is either 0 or $\dim E_\alpha$. Let S be the set of selected complete shells. Then*

$$\Lambda_{\mathcal{A},\mathbf{d}} = \sum_{\alpha,\beta \in S} \sum_{\gamma \notin S} \left\| \Pi_\gamma Q \big|_{E_\alpha \times E_\beta} \right\|_{\mathrm{HS}}^2, \tag{34}$$

which is the complete-shell leakage scalar of Ref. [3].

Proof. At such a vertex, $P_{\mathbf{L}}$ is the orthogonal projector onto the union of the selected shells. Orthogonality of the signed-shell decomposition splits the squared leakage norm into the sum in (34). □

Remark III.21. The complete-shell criterion tests finitely many distinguished vertices of a much larger moduli problem. The function (31) extends that test to every partial-shell configuration with the same finite spectral support.

G. Infinitesimal deformation of the leakage variety

The algebraic zero-locus description also has a concrete first-order deformation theory. We record it in a graph chart, which avoids assigning a spurious codimension to a singular zero set.

Let

$$W = L_\alpha \oplus L_\beta \subset E_\alpha \oplus E_\beta$$

be a zero-leakage point for two distinct signed shells. A tangent vector to the product Grassmannian is represented by

$$X_\alpha \in \mathrm{Hom}(L_\alpha, L_\alpha^\perp), \qquad X_\beta \in \mathrm{Hom}(L_\beta, L_\beta^\perp).$$

Theorem III.22 (Linearized leakage and Zariski tangent space)**.** *In the standard graph trivialization of the leakage bundle, the differential at a zero-leakage point is*

$$\begin{aligned} d\Phi(X_\alpha, X_\beta)(u,v) = (I - P_W)\,[Q(X_\alpha u, v) + Q(u, X_\beta v)] \\ - X_\alpha P_{L_\alpha} Q(u,v) - X_\beta P_{L_\beta} Q(u,v), \end{aligned} \tag{35}$$

for $u \in L_\alpha$, $v \in L_\beta$. Consequently

$$\boxed{T^{\mathrm{Zar}}_{(L_\alpha, L_\beta)} \mathcal{Z}_{\alpha,\beta} = \ker d\Phi.} \tag{36}$$

At a regular point the local codimension equals $\operatorname{rank} d\Phi$.

Proof. Represent nearby planes as graphs of εX_α and εX_β. Differentiate the bilinear leakage

$$(I - P_{W(\varepsilon)}) Q(u + \varepsilon X_\alpha u, v + \varepsilon X_\beta v)$$

at $\varepsilon = 0$. The first line of (35) comes from varying the two arguments, while the second line is the first variation of the target projection. The tangent space to the zero locus of an algebraic section is the kernel of its linearization. At a regular zero, the constant-rank theorem gives the codimension statement. □

Remark III.23. There is no universal formula equating the codimension of the leakage variety with the sum of ranks of the shell projections of Q. Such a formula would require additional transversality and independence hypotheses.

H. Pressure-linear incidence varieties

One may impose the stronger requirement

$$Q(L, L) = 0$$

rather than merely $Q(L, L) \subset L$. For two distinct shells E_α, E_β, same-shell terms already vanish. Hence pressure-linearity of $L_\alpha \oplus L_\beta$ is equivalent to

$$Q(L_\alpha, L_\beta) = 0.$$

Definition III.24. For dimensions d, e, define

$$\mathcal{I}^{d,e}_{\alpha,\beta} := \{(L, M) \in \mathrm{Gr}(d, E_\alpha) \times \mathrm{Gr}(e, E_\beta) : Q(L, M) = 0\}. \tag{37}$$

Proposition III.25 (Cross-shell incidence variety)**.** *The set $\mathcal{I}^{d,e}_{\alpha,\beta}$ is a compact $SO(4)$-invariant real-algebraic subset of the product Grassmannian. Every point parameterizes a pressure-linear exact reduction $L \oplus M$.*

Proof. Use projector coordinates as in Theorem III.18. The condition $Q(L, M) = 0$ is equivalent to the vanishing of the finite family of bilinear coefficients

$$Q(P_L e_i, P_M f_j),$$

hence is algebraic. Stokes invariance is automatic because L and M lie in eigenspaces of A; the same-shell terms vanish by Theorem II.1. □

This incidence geometry separates two questions. Representation theory determines which shell pairs can interact at all; algebraic geometry then determines which partial subspaces annihilate the allowed interaction tensor.

IV. KILLING PRECESSION ON REDUCTION MODULI

A. Killing drift on a signed shell

Let $E_{k,\sigma}$, $k \geq 1$, $\sigma = \pm$, be a real signed curl eigenspace on the round three-sphere of radius R. Its dimension is

$$d_k = (k + 1)(k + 3),$$

and the Stokes operator is scalar:

$$A|_{E_{k,\sigma}} = \kappa_k \mathrm{Id}, \qquad \kappa_k = \frac{k(k + 4)}{R^2}.$$

The full Killing algebra splits as

$$\mathfrak{K} = \mathfrak{K}_+ \oplus \mathfrak{K}_- \simeq \mathfrak{su}(2)_+ \oplus \mathfrak{su}(2)_-.$$

We use the exact shell drift

$$D_{k,\sigma}(m) = a_k \ \mathrm{ad}_{m_\sigma} + b_k \ \mathrm{ad}_{m_{-\sigma}}, \qquad a_k = \frac{k}{k + 2}, \qquad b_k = \frac{k + 4}{k + 2}. \tag{38}$$

Here ad denotes the infinitesimal isometry action on the shell. The two simple factors commute, and $D_{k,\sigma}(m)$ is real skew-adjoint.

The complexification of $E_{k,\sigma}$ is the tensor product

$$V_{j_\sigma} \boxtimes V_{j_{-\sigma}}, \qquad j_\sigma = \frac{k + 2}{2}, \quad j_{-\sigma} = \frac{k}{2}.$$

These are the only representation-theoretic inputs used below.

B. Projector Lax dynamics

Let $L_0 \subset E_{k,\sigma}$ be a real d-dimensional subspace and let $U(t)$ solve

$$\dot{U}(t) = -D(t)U(t), \qquad U(0) = I,$$

where $D(t) = D_{k,\sigma}(m(t))$. Since $D(t)^* = -D(t)$, $U(t)$ is orthogonal.

Define

$$L(t) = U(t)L_0, \qquad P(t) = U(t)P_0U(t)^*.$$

Theorem IV.1 (Exact projector equation)**.** *The orthogonal projector satisfies*

$$\boxed{\dot{P} = [P, D].} \tag{39}$$

If D is constant, then

$$L_D(t) := P(t)D \tag{40}$$

satisfies the Lax equation

$$\boxed{\dot{L}_D = [L_D, D].} \tag{41}$$

Consequently

$$\mathrm{tr}\big(L_D(t)^r\big) = \mathrm{tr}\big(L_D(0)^r\big) \qquad (r \geq 1). \tag{42}$$

Proof. Differentiate $P = UP_0U^*$:

$$\dot{P} = -DP + PD = [P, D].$$

For constant D,

$$\dot{L}_D = [P, D]D = PD^2 - DPD = [PD, D].$$

Trace invariance follows from cyclicity. □

Remark IV.2. The invariants (42) are spectral invariants of one Lax matrix. Their existence does not by itself establish Liouville or Gelfand–Tsetlin complete integrability.

C. Exact geodesic criterion

The physical space of real d-planes is

$$\mathrm{Gr}_{\mathbb{R}}(d, d_k) \simeq O(d_k)/(O(d) \times O(d_k - d)),$$

with its canonical metric. Geodesics in the projector model are generated by horizontal, block-off-diagonal skew matrices [4, 10].

Assume now that D is constant. Relative to the orthogonal splitting

$$E_{k,\sigma} = L_0 \oplus L_0^\perp,$$

write

$$D = \begin{pmatrix} A & -B^* \\ B & C \end{pmatrix}, \qquad A^* = -A, \quad C^* = -C. \tag{43}$$

Theorem IV.3 (Geodesic criterion)**.** *The orbit*

$$P(t) = e^{-tD} P_0 e^{tD}$$

is a canonical Grassmann geodesic if and only if

$$\boxed{CB = BA.} \tag{44}$$

Equivalently, the block-diagonal and block-off-diagonal parts of D commute.

Proof. At $P_0 = \mathrm{diag}(I, 0)$,

$$\dot{P}(0) = \begin{pmatrix} 0 & -B^* \\ -B & 0 \end{pmatrix}.$$

A second differentiation gives

$$\ddot{P}(0) = \begin{pmatrix} -2B^*B & AB^* - B^*C \\ CB - BA & 2BB^* \end{pmatrix}.$$

For the canonical Grassmann metric, the tangential part of an ambient symmetric matrix is precisely its block-off-diagonal part. Thus vanishing covariant acceleration at $t = 0$ is equivalent to $CB = BA$.

If this condition holds, the decomposition $D = D_{\mathrm{diag}} + D_{\mathrm{off}}$ satisfies

$$[D_{\mathrm{diag}}, D_{\mathrm{off}}] = 0, \qquad [D_{\mathrm{diag}}, P_0] = 0.$$

Hence

$$e^{-tD} P_0 e^{tD} = e^{-tD_{\mathrm{off}}} P_0 e^{tD_{\mathrm{off}}},$$

which is the standard Grassmann geodesic generated by the horizontal skew matrix D_{off}. This proves sufficiency as well. □

Corollary IV.4. *A generic constant Killing drift orbit is not a Grassmann geodesic. Geodesicity is an algebraic compatibility condition between the action inside L_0, the action inside $L_0^\perp$, and the mixing block B.*

D. Exact fixed-drift spectrum

Choose pointwise-orthonormal Cartan generators $X_3^\pm$ satisfying the standard round-sphere normalization. By independent rotations in the two simple factors, any fixed Killing momentum may be written

$$m_\sigma = M_\sigma X_3^\sigma, \qquad m_{-\sigma} = M_{-\sigma} X_3^{-\sigma}, \qquad M_\pm \geq 0.$$

On the complexified shell choose the tensor-product weight basis

$$|p, q\rangle, \qquad p = -j_\sigma, -j_\sigma + 1, \ldots, j_\sigma,$$

$$q = -j_{-\sigma}, -j_{-\sigma} + 1, \ldots, j_{-\sigma}.$$

Theorem IV.5 (Exact shell splitting)**.** *The operator*

$$\mathcal{L}_m = \nu A + D_{k,\sigma}(m)$$

has eigenvalues

$$\boxed{\lambda_{p,q} = \nu\kappa_k + \frac{2i}{R}\left(a_k M_\sigma p + b_k M_{-\sigma} q\right).} \tag{45}$$

Proof. The two $SU(2)$ factors commute. The infinitesimal generator X_3^τ has eigenvalues $2ip/R$ in spin j_τ. Apply (38) and add the scalar Stokes part. □

E. Resonance arithmetic and generic simplicity

Two weights (p, q) and (p', q') collide exactly when

$$a_k M_\sigma (p - p') + b_k M_{-\sigma} (q - q') = 0.$$

The differences are integers:

$$r = p - p', \qquad s = q - q',$$

with

$$|r| \leq k + 2, \qquad |s| \leq k.$$

Theorem IV.6 (Finite Diophantine collision criterion)**.** *The spectrum (45) is simple if and only if*

$$a_k M_\sigma r + b_k M_{-\sigma} s \neq 0 \tag{46}$$

for every nonzero admissible integer pair

$$(r, s) \in \mathbb{Z}^2, \qquad |r| \leq k + 2, \quad |s| \leq k.$$

If both M_σ and $M_{-\sigma}$ are nonzero, an irrational ratio

$$\frac{a_k M_\sigma}{b_k M_{-\sigma}}$$

is sufficient for simplicity.

Corollary IV.7 (Exact minimum spacing)**.** *The minimum nonzero frequency spacing is*

$$\boxed{\Delta\omega_k = \frac{2}{R} \min_{\substack{(r,s)\in\mathbb{Z}^2\setminus\{0\} \\ |r|\leq k+2,\ |s|\leq k}} |a_k M_\sigma r + b_k M_{-\sigma} s|.} \tag{47}$$

Thus the minimum spacing can be smaller than either one-axis step because of cross-factor cancellation.

Remark IV.8. Spectral splitting alone does not prove suppression of nonlinear triads. A cascade estimate requires a separate oscillatory-integral theorem and explicit exclusion of resonances satisfying (46) with equality.

F. Special-orthogonal shell transport

The shell drift is a real skew-adjoint representation operator.

Lemma IV.9 (Tracelessness). *For every $m \in \mathfrak{K}$,*

$$\boxed{\operatorname{tr} D_{k,\sigma}(m) = 0.} \tag{48}$$

Proof. Each simple-factor infinitesimal representation of $\mathfrak{su}(2)$ is traceless. The drift is a linear combination of the two commuting simple-factor actions. □

Theorem IV.10 (Determinant-one transport). *Let*

$$\dot{U}(t) = -D_{k,\sigma}(m(t))U(t), \qquad U(0) = I.$$

Then

$$\boxed{U(t) \in SO(d_k), \qquad \det U(t) = 1} \tag{49}$$

for every t. After complexification, $U(t) \in SU(d_k)$.

Proof. Skew-adjointness gives orthogonality. Jacobi's determinant formula and Theorem IV.9 give

$$\frac{\mathrm{d}}{\mathrm{d}t} \log\det U = -\operatorname{tr} D_{k,\sigma}(m(t)) = 0.$$

Since $\det U(0) = 1$, the result follows. □

Corollary IV.11 (No determinant Berry phase). *The determinant line of the full shell has trivial parallel transport. Therefore full-shell precession carries no nontrivial scalar $U(1)$ Berry phase obtained from $\det U$, even though the nonabelian $SO(d_k)$ holonomy can be nontrivial.*

Remark IV.12 (Closed loops). A periodic coefficient $m(T) = m(0)$ does not by itself define holonomy of a closed base loop. If the connection is viewed over $SO(4)$, the reconstructed group path must satisfy $g(T) = g(0)$. Likewise, to speak of a holonomy operator on a particular subspace L_0, the endpoint transport must return that subspace to itself.

G. Curvature without scalar topology

The shell curvature may be written

$$\Omega_{k,\sigma}(m,n) = -\frac{2k}{(k+2)^2}\,\mathrm{ad}_{[m_\sigma, n_\sigma]} + \frac{2(k+4)}{(k+2)^2}\,\mathrm{ad}_{[m_{-\sigma}, n_{-\sigma}]}\,.$$

Every term is traceless, hence

$$\operatorname{tr}\Omega_{k,\sigma} = 0.$$

Thus nonflatness is a genuinely nonabelian phenomenon. It should not be confused with a nonzero first Chern class or with a quantized scalar Chern–Simons charge. Chern–Simons functionals require their own normalization and gauge analysis; in general their real values are not topological integers.

V. KILLING-RESPONSE TRANSFORM AND MINIMAL OBSERVABILITY

A. Killing–Beltrami response identity

Choose pointwise-orthonormal frames X_a^τ in the two simple Killing factors with

$$[X_a^\tau, X_b^\tau] = \frac{2}{R}\varepsilon_{abc}X_c^\tau, \qquad [X_a^+, X_b^-] = 0,$$

and put $K_a^\tau = V_R^{-1/2}X_a^\tau$.

Lemma V.1 (Killing–Beltrami response). *If $K \in E_{0,\tau}$ and $v \in E_{k,\sigma}$, $k \geq 1$, then*

$$\boxed{2Q(K,v) = \left(1 - \frac{2\tau\sigma}{k+2}\right)[K,v].} \tag{50}$$

In particular $Q(K,v) \in E_{k,\sigma}$.

Proof. On an oriented Riemannian three-manifold,

$$\nabla_K v + \nabla_v K = \operatorname{grad}_g\langle K, v\rangle - K \times \operatorname{curl} v - v \times \operatorname{curl} K.$$

With $\lambda = 2\tau/R$ and $\mu = \sigma(k+2)/R$, the non-gradient part is $(\lambda-\mu)K\times v$. Since K is Killing, its flow commutes with curl, so $[K,v] \in E_{k,\sigma}$. Also $\operatorname{curl}(K\times v) = -[K,v]$. After Leray projection,

$$\mathbb{P}(K \times v) = -\mu^{-1}[K,v].$$

Substitution gives (50). □

The representation of the two $SU(2)$ factors on $E_{k,\sigma}$ has spins

$$j_\tau = \frac{k+1+\tau\sigma}{2}. \tag{51}$$

For the L^2-normalized Killing basis,

$$-\sum_{a=1}^{3}\mathcal{L}_{K_a^\tau}^2 = \frac{4}{R^2V_R}j_\tau(j_\tau+1)\mathrm{Id}. \tag{52}$$

B. The six-channel Parseval frame

Set

$$T_a^\tau = Q(K_a^\tau, \cdot).$$

These operators are skew-adjoint and preserve every signed shell.

Define

$$(\mathscr{A}v)_{a,\tau} = \sqrt{2V_R}\,T_a^\tau v$$

on $D(A^{1/2}) \cap \mathcal{H}$.

Theorem V.2 (Casimir factorization)**.** *On $\mathcal{H}$,*

$$\boxed{\mathscr{A}^*\mathscr{A} = A.} \tag{53}$$

Equivalently,

$$\boxed{2V_R \sum_{\tau,a} \|T_a^\tau v\|_2^2 = \langle Av, v\rangle.} \tag{54}$$

Proof. On each signed shell, use (50) and (52). The sum of the two chiral scalar contributions reduces to κ_k. Shell orthogonality then gives the operator identity. □

The same identity may be written in sum-of-squares form as

$$\boxed{A = -2V_R \sum_{\tau=\pm} \sum_{a=1}^{3} (T_a^\tau)^2.} \tag{55}$$

Define the bounded normalized transforms

$$\boxed{\mathscr{R}_a^\tau = \sqrt{2V_R}\, T_a^\tau A^{-1/2} : \mathcal{H} \to L^2_\sigma(TS^3_R).} \tag{56}$$

Theorem V.3 (Parseval Killing-response frame)**.**

$$\boxed{\sum_{\tau=\pm} \sum_{a=1}^{3} (\mathscr{R}_a^\tau)^* \mathscr{R}_a^\tau = I_{\mathcal{H}}.} \tag{57}$$

Hence

$$\|v\|_2^2 = \sum_{\tau,a} \|\mathscr{R}_a^\tau v\|_2^2.$$

Proof. Conjugate (53) by $A^{-1/2}$. □

C. A single chirality is already complete

Let

$$\mathscr{A}_\tau v = \sqrt{2V_R}(T_1^\tau v, T_2^\tau v, T_3^\tau v).$$

Theorem V.4 (Exact chiral frame operator)**.** *On $\mathcal{H}$,*

$$\boxed{\mathscr{A}_\tau^*\mathscr{A}_\tau = \frac{1}{2} A \left(I - \frac{2\tau}{R}\,\mathrm{curl}^{-1}\right),} \tag{58}$$

$$\boxed{\mathscr{S}_\tau := \sum_{a=1}^{3} (\mathscr{R}_a^\tau)^* \mathscr{R}_a^\tau = \frac{1}{2}\left(I - \frac{2\tau}{R}\,\mathrm{curl}^{-1}\right).} \tag{59}$$

Moreover,

$$\boxed{\frac{1}{6} I \le \mathscr{S}_\tau \le \frac{5}{6} I,} \tag{60}$$

and both constants are sharp.

Proof. On $E_{k,\sigma}$, (50) and (52) give

$$\mathscr{A}_\tau^*\mathscr{A}_\tau = \frac{\kappa_k}{2}\left(1 - \frac{2\tau\sigma}{k+2}\right) \mathrm{Id}.$$

Since $\mathrm{curl}^{-1} = \sigma R/(k+2)$ on this shell, (58) follows. Conjugation by $A^{-1/2}$ gives (59). Finally,

$$\frac{2}{k+2} \le \frac{2}{3}, \qquad k \ge 1,$$

so the spectral interval is $[1/6, 5/6]$. Both endpoints occur at $k = 1$. □

Corollary V.5 (Stable three-channel reconstruction)**.** *For either chirality,*

$$\frac{1}{6}\|v\|_2^2 \le \sum_{a=1}^{3} \|\mathscr{R}_a^\tau v\|_2^2 \le \frac{5}{6}\|v\|_2^2. \tag{61}$$

The canonical dual reconstruction is

$$\boxed{v = \mathscr{S}_\tau^{-1} \sum_{a=1}^{3} (\mathscr{R}_a^\tau)^* \mathscr{R}_a^\tau v.} \tag{62}$$

For raw response fields,

$$\boxed{v = -4V_R \left[A\left(I - \frac{2\tau}{R}\,\mathrm{curl}^{-1}\right)\right]^{-1} \sum_{a=1}^{3} T_a^\tau (T_a^\tau v).} \tag{63}$$

D. Two probes determine the entire high-mode field

Fix an orthonormal pair $K_1^\tau, K_2^\tau \in \mathfrak{K}_\tau$ and define

$$\mathscr{A}_{\tau,2} v = \sqrt{2V_R}(T_1^\tau v, T_2^\tau v).$$

Lemma V.6 (Two-axis shell spectrum)**.** *On $E_{k,\sigma}$ set*

$$j = \frac{k+1+\tau\sigma}{2}, \qquad d_\tau = \frac{\kappa_k}{2}\left(1 - \frac{2\tau\sigma}{k+2}\right).$$

In a weight basis of the omitted third generator, the pair frame eigenvalues are

$$\lambda_m^{(2)} = d_\tau \frac{j(j+1) - m^2}{j(j+1)}, \qquad m = -j, \dots, j. \tag{64}$$

Hence

$$\boxed{\lambda_{\min}^{(2)} = \frac{d_\tau}{j+1} > 0.} \tag{65}$$

Proof. The common scalar in (50) cancels in the ratio between the full three-axis Casimir and the two-axis operator. In spin j,

$$J_1^2 + J_2^2 = j(j+1)I - J_3^2.$$

The minimum occurs at $|m| = j$, where the numerator is j. □

Theorem V.7 (Sharp global two-probe bound)**.** *For either chirality and every $v \in D(A^{1/2}) \cap \mathcal{H}$,*

$$\boxed{2V_R\left(\|T_1^\tau v\|_2^2 + \|T_2^\tau v\|_2^2\right) \geq \frac{1}{3R^2}\|v\|_2^2.} \tag{66}$$

The constant is sharp. Thus two fixed same-chirality Killing responses determine the complete non-Killing state and have a bounded inverse on their range in the L^2 topology.

Proof. By shell orthogonality, minimize (65).

If $\tau\sigma = +1$, then

$$j = \frac{k+2}{2}, \qquad \lambda_{\min}^{(2)} = \frac{k^2}{R^2(k+2)}.$$

This is increasing for $k \geq 1$ and equals $1/(3R^2)$ at $k = 1$.

If $\tau\sigma = -1$, then

$$j = \frac{k}{2}, \qquad \lambda_{\min}^{(2)} = \frac{k(k+4)^2}{R^2(k+2)^2},$$

which is larger than $1/(3R^2)$ for all $k \geq 1$. □

Remark V.8. The raw pair is a first-order closed analysis operator, not a bounded map $L^2 \to (L^2)^2$. The estimate (66) states that its inverse from the exact response range to L^2 is bounded. The normalized three- and six-channel transforms are the bounded g-frames.

E. One-probe parity law

Fix $0 \neq K \in \mathfrak{K}_\tau$.

Theorem V.9 (Parity and exact blind-space dimension)**.** *On $E_{k,\sigma}$:*

(i) If k is odd, then

$$T_K = Q(K, \cdot)$$

is invertible.

(ii) If k is even, then T_K has a nontrivial kernel of dimension

$$\boxed{\dim\ker T_K = k + 2 - \tau\sigma.} \tag{67}$$

Thus the dimensions are $k+1$ for $\tau\sigma = +1$ and $k+3$ for $\tau\sigma = -1$.

Proof. By (50), T_K is a nonzero scalar multiple of an $SU(2)_\tau$ infinitesimal generator in spin

$$j_\tau = \frac{k + 1 + \tau\sigma}{2}.$$

After complexification its weights are $-j_\tau, -j_\tau + 1, \ldots, j_\tau$. Zero occurs if and only if j_τ is an integer.

For odd k, both possible spins are half-integers. For even k, both are integers. The zero-weight line in the $SU(2)_\tau$ factor is tensored with the opposite factor of dimension

$$2j_{-\tau} + 1 = k + 2 - \tau\sigma.$$

Complexification preserves the kernel dimension of the real operator. □

Corollary V.10 (One-probe condition number on odd shells)**.** *For odd k,*

$$\boxed{\mathrm{cond}(T_K) = k + 1 + \tau\sigma.} \tag{68}$$

Thus the condition number is $k+2$ for $\tau\sigma = +1$ and k for $\tau\sigma = -1$.

Proof. For half-integer spin j, the largest generator weight has magnitude j and the smallest has magnitude $1/2$. The ratio is $2j$. □

F. Two-probe condition numbers and minimality

Theorem V.11 (Exact pair condition number)**.** *For two orthonormal same-chirality probes on $E_{k,\sigma}$, with $j = (k + 1 + \tau\sigma)/2$,*

$$\boxed{\mathrm{cond}_2 = \begin{cases} \sqrt{j+1}, & j \in \mathbb{Z}, \\ \sqrt{j + 1 - \dfrac{1}{4j}}, & j \in \mathbb{Z} + \dfrac{1}{2}. \end{cases}} \tag{69}$$

Hence the pair condition number is $O(k^{1/2})$, whereas one-channel inversion on odd shells is $O(k)$.

Proof. The squared singular values are proportional to $j(j+1) - m^2$ by (64). The minimum is j. For integer j, the maximum is $j(j+1)$ at $m = 0$. For half-integer j, it is $j(j+1) - 1/4$ at $|m| = 1/2$. □

Corollary V.12 (Minimal universal probe count)**.** *Within either fixed chiral Killing factor:*

(i) two linearly independent fixed probes are sufficient to distinguish every non-Killing state;

(ii) one fixed probe is never sufficient uniformly over all shells because every even shell has a blind subspace;

(iii) on a prescribed odd shell the minimal number is one, while on a prescribed even shell it is two.

G. Erasure hierarchy

The reconstruction structure can be summarized as follows.

Theorem V.13 (Chiral erasure hierarchy)**.** *Fix $E_{k,\sigma}$ and one Killing chirality.*

(i) Three orthonormal response channels form a tight operator frame on the shell.

(ii) Erasing any one channel leaves a frame with lower eigenvalue $d_\tau/(j_\tau+1)$.

(iii) Erasing two channels leaves an injective one-channel transform if and only if k is odd.

(iv) On even shells, every one-channel transform has the blind-space dimension (67).

The result gives exact loss tolerance of the Killing-response representation.

VI. CASIMIR RESPONSE GEOMETRY

A. From lossless response fields to Gram compression

The frame results above use the response fields themselves. We now compress those six fields to their mutual inner products. This passage is useful but not lossless. It produces finite-dimensional isometry covariants and exact spectral statistics, while forgetting part of the common phase information. This hierarchy should be kept explicit:

$$\boxed{v \longmapsto \{Q(K_a^\tau, v)\}_{\tau,a} \longmapsto \mathscr{M}_v \longmapsto \{\mathscr{M}_v^{[s]}\}_s \longmapsto \Phi_v(\theta).}$$

The first arrow is injective by the two-probe theorem. Later we show that the Gram-level map cannot be injective on sufficiently high individual shells.

B. The Casimir response tensor

Definition VI.1 (Response operator and response tensor)**.** For a divergence-free field u in the form domain of A, define

$$\mathscr{B}_u : \mathfrak{K} \longrightarrow L^2_\sigma(TS^3_R), \qquad \mathscr{B}_u X := \sqrt{2V_R}\, Q(X,u). \tag{70}$$

Its Casimir response tensor is

$$\boxed{\mathscr{M}_u := \mathscr{B}_u^* \mathscr{B}_u} \tag{71}$$

or, equivalently,

$$\mathscr{M}_u(X,Y) = 2V_R \langle Q(X,u), Q(Y,u)\rangle_{L^2}. \tag{72}$$

The tensor is positive semidefinite and quadratic in u:

$$\mathscr{M}_{cu} = c^2 \mathscr{M}_u.$$

Theorem VI.2 (Trace identity and exact dissipation)**.** *For every divergence-free u in the form domain of A,*

$$\boxed{\operatorname{tr}_{\mathfrak{K}} \mathscr{M}_u = \langle Au, u\rangle = 2\| \operatorname{Def} u\|_2^2.} \tag{73}$$

Hence the viscous energy dissipation satisfies

$$\boxed{\varepsilon_\nu(u) := 2\nu \| \operatorname{Def} u\|_2^2 = \nu \operatorname{tr} \mathscr{M}_u.} \tag{74}$$

Proof. The quadratic Casimir identity for the six L^2-normalized Killing responses is

$$A = -2V_R \sum_{\tau=\pm} \sum_{a=1}^{3} T_a^\tau T_a^\tau, \qquad T_a^\tau := Q(K_a^\tau, \cdot), \tag{75}$$

where each T_a^τ is skew-adjoint. Therefore

$$\langle Au, u\rangle = 2V_R \sum_{\tau,a} \|Q(K_a^\tau, u)\|_2^2 = \operatorname{tr} \mathscr{M}_u.$$

For the deformation-tensor viscosity, $\langle Au, u\rangle = 2\| \operatorname{Def} u\|_2^2$. □

Remark VI.3. The response tensor is a global L^2 object. Equation (73) does not, by itself, imply spatial localization, microlocal anisotropy, or a preferred geometric axis in physical space.

C. Isometry covariance and the response ellipsoid

Let $g \in SO(4)$ act unitarily on vector fields by push-forward. The induced action on $\mathfrak{K}$ is the adjoint action.

Theorem VI.4 (Isometry covariance)**.** *For every $g \in SO(4)$,*

$$\mathscr{M}_{g_* u} = \operatorname{Ad}_g \, \mathscr{M}_u \, \operatorname{Ad}_{g^{-1}}. \tag{76}$$

Consequently the ordered eigenvalues

$$\lambda_1(u) \geq \cdots \geq \lambda_6(u) \geq 0 \tag{77}$$

are invariants of the physical isometry orbit of u.

Proof. Isometry equivariance of the Levi-Civita connection and Leray projection gives

$$Q(X, g_* u) = g_* Q(\operatorname{Ad}_{g^{-1}} X, u).$$

Since g_* is unitary in L^2, substitution into (72) gives (76). □

Definition VI.5 (Response ellipsoid)**.** The image of the unit sphere in $\mathfrak{K}$ under $\mathscr{B}_u$ is the response ellipsoid in $\operatorname{ran} \mathscr{B}_u$. Its principal semiaxis lengths are

$$\sqrt{\lambda_1(u)}, \ldots, \sqrt{\lambda_6(u)}, \tag{78}$$

with zero axes allowed.

Remark VI.6. The eigenvalues themselves are *squared* semiaxis lengths. This distinction is important when interpreting the geometry.

Corollary VI.7 (Extremal-response principle)**.** *For every u,*

$$\frac{1}{6}\langle Au,u\rangle \le \lambda_1(u) \le \langle Au,u\rangle. \tag{79}$$

Hence along any sequence u_n,

$$\|\operatorname{Def} u_n\|_2 \longrightarrow \infty \quad\Longleftrightarrow\quad \lambda_1(u_n)\longrightarrow\infty. \tag{80}$$

Proof. For a positive semidefinite 6×6 operator,

$$\lambda_1 \le \sum_i \lambda_i \le 6\lambda_1.$$

Use Theorem VI.2. □

Remark VI.8. Equation (80) does not say that dissipation growth must become anisotropic. All six eigenvalues may grow simultaneously. A separate normalized spectral diagnostic is required to discuss response polarization.

D. Spectral additivity

Write

$$u=\sum_{k,\sigma} u_{k,\sigma},\qquad u_{k,\sigma}\in E_{k,\sigma}.$$

Theorem VI.9 (Spectral additivity of the response tensor)**.** *For every u in the form domain of A,*

$$\boxed{\mathscr{M}_u=\sum_{k,\sigma}\mathscr{M}_{u_{k,\sigma}}} \tag{81}$$

with convergence in the finite-dimensional operator norm on $\mathfrak{K}$ whenever $\langle Au,u\rangle<\infty$.

Proof. By (50),

$$Q(X,u_{k,\sigma})\in E_{k,\sigma}$$

for every $X\in\mathfrak{K}$. Distinct signed shells are L^2-orthogonal. Hence for $X,Y\in\mathfrak{K}$,

$$\langle Q(X,u),Q(Y,u)\rangle=\sum_{k,\sigma}\langle Q(X,u_{k,\sigma}),Q(Y,u_{k,\sigma})\rangle.$$

This is (81). The trace of the positive series is $\langle Au,u\rangle$, so convergence follows in finite dimension. □

This identity says that shell phases can affect the shape of each $\mathscr{M}_{u_{k,\sigma}}$, but there are no cross-shell interference terms in the response tensor itself.

E. Exact chiral splitting

Let $P_\tau:\mathfrak{K}\to\mathfrak{K}_\tau$ be the orthogonal projections and define the two chiral response traces

$$D_\tau(u):=\operatorname{tr}_{\mathfrak{K}_\tau}\mathscr{M}_u=2V_R\sum_{a=1}^{3}\|Q(K_a^\tau,u)\|_2^2. \tag{82}$$

Then

$$D_+(u)+D_-(u)=\langle Au,u\rangle. \tag{83}$$

Theorem VI.10 (Pure-shell chiral partition)**.** *Let $u\in E_{k,\sigma}$ with $k\ge 1$. Then*

$$\boxed{D_\tau(u)=\frac{\kappa_k}{2}\left(1-\frac{2\tau\sigma}{k+2}\right)\|u\|_2^2.} \tag{84}$$

Equivalently,

$$D_\sigma(u)=\frac{k}{2(k+2)}\kappa_k\|u\|_2^2, \tag{85}$$

$$D_{-\sigma}(u)=\frac{k+4}{2(k+2)}\kappa_k\|u\|_2^2. \tag{86}$$

Thus the opposite Killing chirality carries the larger partial response trace, and the relative imbalance is

$$\frac{D_{-\sigma}(u)-D_\sigma(u)}{D_{-\sigma}(u)+D_\sigma(u)}=\frac{2}{k+2}. \tag{87}$$

Proof. Set

$$c_\tau(k,\sigma):=\frac{k+2-2\tau\sigma}{k+2}.$$

By (50),

$$Q(K_a^\tau,u)=\frac{c_\tau}{2}\mathcal{L}_{K_a^\tau}u.$$

Using the factor Casimir (52),

$$\begin{aligned}D_\tau(u)&=2V_R\frac{c_\tau^2}{4}\sum_{a=1}^{3}\|\mathcal{L}_{K_a^\tau}u\|_2^2\\&=\frac{2}{R^2}c_\tau^2 j_\tau(j_\tau+1)\|u\|_2^2.\end{aligned}$$

If $\tau\sigma=1$, then $c_\tau=k/(k+2)$ and $j_\tau=(k+2)/2$, giving (85). If $\tau\sigma=-1$, then $c_\tau=(k+4)/(k+2)$ and $j_\tau=k/2$, giving (86). Formula (84) combines the two cases. □

Remark VI.11 (High-frequency balance without connection flattening). The relative trace bias (87) tends to zero as $k\to\infty$. This does not contradict persistence of nonzero operator-norm curvature in the shell-precession connection: response-trace balance and connection flatness are different operator statements.

Theorem VI.12 (Global chiral trace identity)**.** *Let u be divergence-free with $\langle Au, u\rangle < \infty$. Then*

$$D_\tau(u) = \sum_{k\geq 1,\ \sigma=\pm} \frac{\kappa_k}{2}\left(1 - \frac{2\tau\sigma}{k+2}\right)\|u_{k,\sigma}\|_2^2, \tag{88}$$

$$D_-(u) - D_+(u) = \frac{2}{R}\langle A\,\mathrm{curl}^{-1}\,u, u\rangle. \tag{89}$$

The Killing component does not contribute because $\kappa_0 = 0$.

Proof. Apply Theorems VI.9 and VI.10. For the difference,

$$D_- - D_+ = 2\sum_{k,\sigma}\sigma\frac{\kappa_k}{k+2}\|u_{k,\sigma}\|_2^2.$$

Since

$$\frac{1}{\mu_{k,\sigma}} = \frac{\sigma R}{k+2},$$

the right-hand side equals

$$\frac{2}{R}\sum_{k,\sigma}\frac{\kappa_k}{\mu_{k,\sigma}}\|u_{k,\sigma}\|_2^2 = \frac{2}{R}\langle A\,\mathrm{curl}^{-1}\,u, u\rangle.$$

□

Definition VI.13 (Chiral response index)**.** For u with $\langle Au, u\rangle > 0$, define

$$\Xi(u) := \frac{D_-(u) - D_+(u)}{D_-(u) + D_+(u)}. \tag{90}$$

Corollary VI.14 (Sharp universal chiral bound)**.** *For every non-Killing u in the form domain of A,*

$$\boxed{|\Xi(u)| \leq \frac{2}{3}.} \tag{91}$$

The constant $2/3$ is sharp. For $u \in E_{k,\sigma}$,

$$\boxed{\Xi(u) = \frac{2\sigma}{k+2}.} \tag{92}$$

Proof. By (88), Ξ is a convex weighted average of the numbers $2\sigma/(k+2)$ with nonnegative weights $\kappa_k\|u_{k,\sigma}\|_2^2$. Since $k \geq 1$ on the dissipative sector, their absolute values are at most $2/3$. Equality is attained on $E_{1,+}$ or $E_{1,-}$. □

F. The cross-chiral block

Relative to $\mathfrak{K} = \mathfrak{K}_+ \oplus \mathfrak{K}_-$, write

$$\mathscr{M}_u = \begin{pmatrix} M_u^{++} & M_u^{+-} \\ M_u^{-+} & M_u^{--} \end{pmatrix}. \tag{93}$$

The diagonal traces are $D_\pm(u)$.

Proposition VI.15 (Universal cross-block bound)**.** *For every u,*

$$\boxed{\|M_u^{+-}\|_{\mathrm{HS}} \leq \sqrt{D_+(u)D_-(u)}.} \tag{94}$$

For a pure shell $u \in E_{k,\sigma}$,

$$\|M_u^{+-}\|_{\mathrm{HS}} \leq \frac{\sqrt{k(k+4)}}{2(k+2)}\kappa_k\|u\|_2^2. \tag{95}$$

Proof. Split the response map as

$$\mathscr{B}_u = (\mathscr{B}_{u,+}, \mathscr{B}_{u,-})$$

on the two Killing factors. Then

$$M_u^{+-} = \mathscr{B}_{u,+}^*\mathscr{B}_{u,-}.$$

Therefore

$$\|M_u^{+-}\|_{\mathrm{HS}} \leq \|\mathscr{B}_{u,+}\|_{\mathrm{HS}}\|\mathscr{B}_{u,-}\|_{\mathrm{HS}} = \sqrt{D_+(u)D_-(u)}.$$

Insert Theorem VI.10 for (95). □

Remark VI.16. The cross block generally need not vanish for a fixed pure-shell vector. Thus it is incorrect to infer from shell irreducibility that the full 6×6 response tensor is block diagonal or scalar on each $\mathfrak{K}_\tau$ for every u. Only the two block traces are fixed pointwise; full block isotropy emerges after an invariant ensemble average.

G. Uniform pure-shell ensemble

Fix a shell $E = E_{k,\sigma}$ of real dimension d_k and let u be uniformly distributed on its L^2 unit sphere.

Theorem VI.17 (Ensemble-averaged response tensor)**.** *Let*

$$d_\tau^{(k,\sigma)} := \frac{\kappa_k}{2}\left(1 - \frac{2\tau\sigma}{k+2}\right). \tag{96}$$

Then

$$\mathbb{E}\,M_u^{\tau\tau} = \frac{d_\tau^{(k,\sigma)}}{3}\mathrm{Id}_{\mathfrak{K}_\tau}, \tag{97}$$

$$\mathbb{E}\,M_u^{+-} = 0, \tag{98}$$

$$\boxed{\mathbb{E}\,\mathscr{M}_u = \begin{pmatrix} \frac{d_+^{(k,\sigma)}}{3}I_3 & 0 \\ 0 & \frac{d_-^{(k,\sigma)}}{3}I_3 \end{pmatrix}.} \tag{99}$$

Proof. The uniform measure on the unit sphere of E is invariant under the full orthogonal group of E. For fixed linear operators S, T on E,

$$\mathbb{E}\,\langle Su, Tu\rangle = \frac{1}{d_k}\,\mathrm{tr}(S^*T).$$

Within one simple factor, the trace pairing of the three representation generators is an invariant bilinear form

and therefore proportional to δ_{ab}. Its total trace is already fixed by Theorem VI.10, giving (97).

For opposite simple factors, the representation is a tensor product. Each Lie generator is traceless on its irreducible spin factor, so

$$\operatorname{tr}\left[(\mathcal{L}_{K_a^+})^*\mathcal{L}_{K_b^-}\right] = 0.$$

Equation (VI G) then gives (98). □

Remark VI.18. The deterministic shell bias survives ensemble averaging through unequal block traces, while directional anisotropy inside each $\mathfrak{K}_\tau$ disappears in the mean.

H. Response polarization diagnostics

The Casimir identity determines only the sum of the six eigenvalues. To distinguish isotropic from concentrated response spectra, define

$$p_i(u) := \frac{\lambda_i(u)}{\operatorname{tr}\mathscr{M}_u}, \qquad \sum_{i=1}^{6} p_i = 1, \tag{100}$$

whenever $\operatorname{tr}\mathscr{M}_u > 0$.

Two useful invariants are

$$\mathcal{P}_\infty(u) := 6p_1(u) = \frac{6\lambda_1(u)}{\operatorname{tr}\mathscr{M}_u}, \tag{101}$$

$$r_{\mathrm{eff}}(u) := \frac{1}{\sum_i p_i(u)^2} = \frac{(\operatorname{tr}\mathscr{M}_u)^2}{\operatorname{tr}(\mathscr{M}_u^2)}. \tag{102}$$

Proposition VI.19 (Response-rank bounds)**.** *For every non-Killing u,*

$$1 \le \mathcal{P}_\infty(u) \le 6, \qquad 1 \le r_{\mathrm{eff}}(u) \le 6. \tag{103}$$

Both quantities are invariant under $SO(4)$ and under nonzero scalar multiplication of u.

Proof. The statements are the standard inequalities for a probability vector $(p_1,\dots,p_6)$, together with Theorem VI.4 and $\mathscr{M}_{cu} = c^2\mathscr{M}_u$. □

Remark VI.20. A divergence of $\operatorname{tr}\mathscr{M}_u$ says nothing by itself about the limiting values of $\mathcal{P}_\infty$ or r_{eff}. Therefore the Casimir identity forces at least one response to become large if dissipation becomes large, but it does not force the normalized response spectrum to polarize.

I. Relation to helicity

The chiral response index (90) is not the normalized fluid helicity. Indeed,

$$\mathscr{H}(u) = \langle u, \operatorname{curl} u\rangle = \sum_{k,\sigma} \sigma \frac{k+2}{R}\|u_{k,\sigma}\|_2^2, \tag{104}$$

whereas (89) uses the inverse-curl weight $\kappa_k/\mu_{k,\sigma}$.

The two quantities emphasize opposite spectral ends: helicity weights high curl frequency more strongly, while the response difference contains the factor $1/(k+2)$ relative to the Stokes dissipation weight.

Proposition VI.21 (Killing precession is transparent to weighted chirality)**.** *Let $D(m) = 2Q(m,\cdot)$ be the shellwise Killing-precession operator. Then*

$$\langle D(m)v, A\operatorname{curl}^{-1} v\rangle = 0 \tag{105}$$

for every sufficiently regular v. Equivalently, the quadratic form on the right side of (89) receives no instantaneous contribution from Killing precession.

Proof. The operator $D(m)$ is skew-adjoint and commutes with both A and curl on the signed-shell decomposition. Therefore $A\operatorname{curl}^{-1}$ is self-adjoint and commutes with $D(m)$, so

$$\begin{aligned}\langle Dv, A\operatorname{curl}^{-1} v\rangle &= -\langle v, DA\operatorname{curl}^{-1} v\rangle \\ &= -\langle v, A\operatorname{curl}^{-1} Dv\rangle.\end{aligned}$$

which is the negative of the original real scalar. □

Remark VI.22. This transparency does not imply that the nonlinear high–high term $Q(v,v)$ preserves shellwise $+/-$ energy equality. In particular, zero initial helicity does not force pairwise equality $\|v_{k,+}(t)\|_2 = \|v_{k,-}(t)\|_2$ for later times.

VII. STOKES RESPONSE MOMENTS AND FINITE-BAND TOMOGRAPHY

A. The Casimir response hierarchy

Because A is invertible on $\mathfrak{K}^\perp$, negative fractional powers are well defined there.

Definition VII.1 (Response hierarchy)**.** Let $s \ge -1$ and suppose $v \in \mathfrak{K}^\perp$ belongs to the domain of $A^{(s+1)/2}$. For $X, Y \in \mathfrak{K}$, define

$$\boxed{\mathscr{M}_v^{[s]}(X,Y) := 2V_R\left\langle A^{s/2}Q(X,v), A^{s/2}Q(Y,v)\right\rangle.} \tag{106}$$

For $s=-1$, $A^{-1/2}$ is taken on $\mathfrak{K}^\perp$.

Each $\mathscr{M}_v^{[s]}$ is a positive semidefinite symmetric operator on the six-dimensional Killing algebra.

Theorem VII.2 (Exact trace ladder)**.** *For every $s \ge -1$,*

$$\boxed{\operatorname{tr}_{\mathfrak{K}}\mathscr{M}_v^{[s]} = \langle A^{s+1}v, v\rangle = \left\|A^{(s+1)/2}v\right\|_2^2.} \tag{107}$$

In particular,

$$\operatorname{tr}\mathscr{M}_v^{[-1]} = \|v\|_2^2, \tag{108}$$

$$\operatorname{tr}\mathscr{M}_v^{[0]} = \langle Av, v\rangle = 2\|\operatorname{Def} v\|_2^2. \tag{109}$$

Proof. Since every T_a^τ commutes with A and is skew-adjoint,

$$\begin{aligned}\operatorname{tr}\mathscr{M}_v^{[s]} &= 2V_R\sum_{\tau,a}\langle T_a^\tau v, A^s T_a^\tau v\rangle \\ &= -2V_R\sum_{\tau,a}\left\langle v, A^s (T_a^\tau)^2 v\right\rangle .\end{aligned}$$

Use (55). □

Theorem VII.3 (Spectral additivity)**.** *If*

$$v=\sum_{k\ge 1,\ \sigma=\pm} v_{k,\sigma},$$

then

$$\boxed{\mathscr{M}_v^{[s]}=\sum_{k,\sigma}\mathscr{M}_{v_{k,\sigma}}^{[s]}.} \tag{110}$$

Whenever the trace in (107) is finite, the series converges in operator norm on $\mathfrak{K}$.

Proof. Each $Q(X, v_{k,\sigma})$ remains in $E_{k,\sigma}$, and distinct signed shells are orthogonal. Hence all cross-shell Gram terms vanish. The summands are positive semidefinite and their traces sum to (107); finite-dimensional monotone convergence then gives operator-norm convergence. □

B. The Stokes moment sequence

Define the total energy at Stokes level k by

$$E_k := \|v_{k,+}\|_2^2 + \|v_{k,-}\|_2^2. \tag{111}$$

Definition VII.4 (Casimir response moments)**.** For integers $n \ge 0$, set

$$\boxed{m_n(v) := \operatorname{tr}\mathscr{M}_v^{[n-1]} = \langle A^n v, v\rangle.} \tag{112}$$

By the signed-shell decomposition,

$$\boxed{m_n=\sum_{k\ge 1}\kappa_k^n E_k.} \tag{113}$$

Thus (m_n) is the moment sequence of the positive discrete measure

$$\mu_v := \sum_{k\ge 1} E_k\,\delta_{\kappa_k}. \tag{114}$$

Theorem VII.5 (Hankel positivity and exact spectral rank)**.** *For $N \ge 0$, define the Hankel matrix*

$$H_N(v) := [m_{i+j}(v)]_{i,j=0}^N. \tag{115}$$

Then $H_N(v)$ is positive semidefinite. If v has exactly r active Stokes levels $k_1, \dots, k_r$, then

$$\boxed{\operatorname{rank} H_N(v) = \min\{N+1, r\}.} \tag{116}$$

Consequently,

$$\det H_{r-1} > 0, \qquad \det H_r = 0. \tag{117}$$

Proof. For $c = (c_0, \dots, c_N)$ and $p(\lambda) = \sum_{j=0}^N c_j \lambda^j$,

$$\begin{aligned}c^{\mathsf{T}} H_N c &= \sum_{i,j} c_i c_j m_{i+j} \\ &= \sum_k E_k |p(\kappa_k)|^2 \ge 0.\end{aligned}$$

If the support consists of r distinct nodes, write

$$H_N = V_N \operatorname{diag}(E_{k_1}, \dots, E_{k_r}) V_N^{\mathsf{T}},$$

where $(V_N)_{ij} = \kappa_{k_j}^i$. The Vandermonde matrix has rank $\min\{N+1, r\}$. □

Remark VII.6. Hankel rank is used broadly in finite moment problems and realization theory [13]. Here the point is not the linear-algebra fact itself, but that the required moments are exact traces of Killing-response tensors.

C. Log-convexity and an exact spectral-spread identity

Theorem VII.7 (Log-convex response moments)**.** *For every $n \ge 1$ for which the moments are finite,*

$$\boxed{m_n^2 \le m_{n-1} m_{n+1}.} \tag{118}$$

If $v \ne 0$, equality holds for some $n \ge 1$ if and only if the non-Killing field is supported on one Stokes level

$$E_{k,+} \oplus E_{k,-}$$

for a single k.

Proof. Apply Cauchy–Schwarz to $A^{(n-1)/2}v$ and $A^{(n+1)/2}v$. Equality requires these two vectors to be linearly dependent. Since A is self-adjoint with distinct eigenvalues κ_k for distinct levels, this occurs exactly on one Stokes eigenspace. □

For the first inequality one can write the defect exactly.

Theorem VII.8 (Spectral-variance identity)**.** *For every $v \in D(A)$,*

$$\boxed{m_0 m_2 - m_1^2 = \frac{1}{2}\sum_{k,\ell\ge 1} E_k E_\ell (\kappa_k - \kappa_\ell)^2.} \tag{119}$$

Equivalently,

$$\frac{m_2}{m_0} - \left(\frac{m_1}{m_0}\right)^2 = \operatorname{Var}_p(\kappa), \qquad p_k = \frac{E_k}{m_0}. \tag{120}$$

Thus three response traces detect exact Stokes monochromaticity.

Proof. Expand the right side:

$$\frac{1}{2}\sum_{k,\ell} E_k E_\ell(\kappa_k^2+\kappa_\ell^2-2\kappa_k\kappa_\ell) = m_0 m_2 - m_1^2.$$

□

Corollary VII.9 (Monotone effective Stokes frequency)**.** *Assume all moments involved are finite and define*

$$\Omega_n(v) := \frac{m_{n+1}(v)}{m_n(v)}. \tag{121}$$

Then Ω_n is nondecreasing in n. For finite spectral support,

$$\lim_{n\to\infty} \Omega_n = \max\{\kappa_k : E_k > 0\}. \tag{122}$$

Proof. Monotonicity is (118). For finite support, divide numerator and denominator by the largest active eigenvalue to the power n. □

D. Chiral moment sequence

Let

$$D_\tau^{[s]}(v) := \mathrm{tr}_{\mathfrak{K}_\tau}\, \mathscr{M}_v^{[s]}. \tag{123}$$

The signed-shell Killing–Beltrami algebra gives the pure-shell formula

$$D_\tau^{[s]}(v_{k,\sigma}) = \frac{\kappa_k^{s+1}}{2}\left(1-\frac{2\tau\sigma}{k+2}\right)\|v_{k,\sigma}\|_2^2. \tag{124}$$

Theorem VII.10 (Higher chiral trace identity)**.** *For every $s \geq -1$,*

$$D_+^{[s]}(v) + D_-^{[s]}(v) = \langle A^{s+1}v, v\rangle, \tag{125}$$

$$\boxed{D_-^{[s]}(v) - D_+^{[s]}(v) = \frac{2}{R}\langle A^{s+1}\,\mathrm{curl}^{-1}\,v, v\rangle.} \tag{126}$$

Proof. Sum (124) over signed shells and use

$$\mu_{k,\sigma}^{-1} = \sigma\frac{R}{k+2}.$$

□

Definition VII.11 (Signed response moments)**.** For $n \geq 0$, define

$$\begin{aligned} h_n(v) &:= \frac{R}{2}\left(D_-^{[n-1]}(v) - D_+^{[n-1]}(v)\right) \\ &= \langle A^n\,\mathrm{curl}^{-1}\,v, v\rangle. \end{aligned} \tag{127}$$

Writing

$$E_{k,\pm} := \|v_{k,\pm}\|_2^2,$$

one obtains

$$\boxed{h_n = \sum_{k\geq 1}\kappa_k^n \frac{R}{k+2}\big(E_{k,+} - E_{k,-}\big).} \tag{128}$$

E. Finite-band spectral tomography

We now show that the two moment sequences recover the complete signed-shell energy spectrum.

Theorem VII.12 (Exact finite-band tomography)**.** *Suppose v has nonzero energy on exactly r distinct Stokes levels*

$$k_1, \dots, k_r.$$

Then the finite data

$$m_0, m_1, \dots, m_{2r}, \qquad h_0, h_1, \dots, h_{r-1} \tag{129}$$

determine uniquely

$$\{k_j, E_{k_j,+}, E_{k_j,-}\}_{j=1}^r.$$

Proof. By Theorem VII.5, r is the unique integer for which

$$\det H_{r-1} > 0, \qquad \det H_r = 0.$$

Let

$$p(z) = \prod_{j=1}^r (z-\kappa_{k_j}) = z^r + c_{r-1}z^{r-1} + \cdots + c_0.$$

Since $p(\kappa_{k_j}) = 0$,

$$\sum_{\ell=0}^r c_\ell m_{n+\ell} = 0, \qquad n = 0, \dots, r-1, \qquad c_r = 1. \tag{130}$$

The coefficient vector $(c_0, \dots, c_{r-1})$ therefore solves

$$H_{r-1}\begin{pmatrix} c_0 \\ \vdots \\ c_{r-1}\end{pmatrix} = -\begin{pmatrix} m_r \\ \vdots \\ m_{2r-1}\end{pmatrix}. \tag{131}$$

The matrix is positive definite, so the solution is unique. The roots of p are exactly the active Stokes eigenvalues. Since $\kappa_k = k(k+4)/R^2$ is strictly increasing in k, the levels k_j are recovered uniquely.

Now solve the Vandermonde system

$$m_n = \sum_{j=1}^r \kappa_{k_j}^n E_{k_j}, \qquad n = 0, \dots, r-1, \tag{132}$$

for

$$E_{k_j} = E_{k_j,+} + E_{k_j,-}.$$

Likewise solve

$$h_n = \sum_{j=1}^r \kappa_{k_j}^n \delta_j, \qquad \delta_j := \frac{R}{k_j+2}(E_{k_j,+} - E_{k_j,-}), \tag{133}$$

for $n = 0, \dots, r-1$. Finally,

$$\boxed{E_{k_j,\pm} = \frac{1}{2}\left[E_{k_j} \pm \frac{k_j+2}{R}\delta_j\right].} \tag{134}$$

□

Remark VII.13. The reconstruction is an exact finite moment problem and can be implemented by standard Prony/Hankel techniques. Numerical conditioning is a separate issue; Hankel moment matrices are well known to become ill-conditioned at high order [13].

F. Complete monotonicity under Stokes relaxation

Consider the unforced linear Stokes equation on $\mathfrak{K}^{\perp}$,

$$\partial_t v + \nu A v = 0. \tag{135}$$

Theorem VII.14 (Loewner-complete monotonicity)**.** *For every $s \geq -1$,*

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{M}^{[s]}_{v(t)} = -2\nu\mathscr{M}^{[s+1]}_{v(t)}.} \tag{136}$$

Consequently, for every integer $q \geq 0$,

$$\boxed{(-1)^q \frac{\mathrm{d}^q}{\mathrm{d}t^q}\mathscr{M}^{[s]}_{v(t)} = (2\nu)^q \mathscr{M}^{[s+q]}_{v(t)} \succeq 0.} \tag{137}$$

Thus $t \mapsto \mathscr{M}^{[s]}_{v(t)}$ is completely monotone in the Loewner order.

Proof. Since every Killing response commutes with A,

$$\frac{\mathrm{d}}{\mathrm{d}t}A^{s/2}Q(X,v) = -\nu A^{s/2+1}Q(X,v).$$

Differentiate the two factors in (106). The two equal terms combine to give (136); iteration gives (137). □

Corollary VII.15 (Scalar moment derivatives)**.** *Under (135),*

$$\frac{\mathrm{d}}{\mathrm{d}t}m_n(t) = -2\nu m_{n+1}(t), \tag{138}$$

and

$$(-1)^q \frac{\mathrm{d}^q}{\mathrm{d}t^q}m_n(t) = (2\nu)^q m_{n+q}(t). \tag{139}$$

G. Two-signal Stokes spectroscopy

Let

$$\mathcal{E}(t) := \|v(t)\|_2^2, \tag{140}$$
$$\mathcal{C}(t) := \langle \mathrm{curl}^{-1}\, v(t), v(t)\rangle. \tag{141}$$

Theorem VII.16 (Relaxation tomography)**.** *For the Stokes evolution (135),*

$$\mathcal{E}(t) = \sum_{k\geq 1} E_k(0)e^{-2\nu\kappa_k t}, \tag{142}$$

$$\mathcal{C}(t) = \sum_{k\geq 1} \frac{R}{k+2}\big(E_{k,+}(0) - E_{k,-}(0)\big)e^{-2\nu\kappa_k t}. \tag{143}$$

Moreover,

$$\boxed{m_n(0) = \frac{(-1)^n}{(2\nu)^n}\mathcal{E}^{(n)}(0),} \tag{144}$$

$$\boxed{h_n(0) = \frac{(-1)^n}{(2\nu)^n}\mathcal{C}^{(n)}(0).} \tag{145}$$

Hence, for a finite-band state with r active Stokes levels, the derivatives

$$\mathcal{E}^{(0)}(0), \ldots, \mathcal{E}^{(2r)}(0), \qquad \mathcal{C}^{(0)}(0), \ldots, \mathcal{C}^{(r-1)}(0)$$

determine the complete signed-shell energy distribution.

Proof. The Stokes semigroup multiplies each shell by $e^{-\nu\kappa_k t}$. Squaring gives (142)–(143). Differentiate at $t = 0$ and use Theorem VII.12. □

Remark VII.17 (Relation to Hankel methods in fluid mechanics). Time-delay Hankel matrices are widely used for data-driven fluid decompositions and are closely related to space-time POD [14]. The two scalar curves above solve a different inverse problem: in the exact round-sphere Stokes dynamics their exponential nodes are fixed geometric eigenvalues, and their amplitudes are the signed shell energies.

H. The nonlinear tensor balance

Return to the high-mode Navier–Stokes equation on an affine Killing leaf,

$$\partial_t v + \nu A v + Q(v,v) + D(m)v = F_{\perp}, \tag{146}$$

where $D(m)$ is skew-adjoint and commutes with A.

For $s \geq -1$ and high-mode fields v, w, define the symmetric cross-response tensor

$$\mathscr{C}_s[v;w](X,Y) := 2V_R\Big[\langle A^{s/2}Q(X,w), A^{s/2}Q(Y,v)\rangle + \langle A^{s/2}Q(X,v), A^{s/2}Q(Y,w)\rangle\Big]. \tag{147}$$

Lemma VII.18 (Trace polarization)**.** *For all sufficiently regular v, w,*

$$\boxed{\mathrm{tr}\,\mathscr{C}_s[v;w] = 2\langle A^{s+1}v, w\rangle.} \tag{148}$$

Proof. Polarize the proof of Theorem VII.2. □

Theorem VII.19 (Exact nonlinear response balance)**.** *Along (146),*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{M}^{[s]}_v = -2\nu\mathscr{M}^{[s+1]}_v - \mathscr{C}_s[v;Q(v,v)] - \mathscr{C}_s[v;D(m)v] + \mathscr{C}_s[v;F_{\perp}]. \tag{149}$$

Taking the trace gives, with $n = s + 1 \geq 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle A^n v, v\rangle = -2\nu\langle A^{n+1}v, v\rangle - 2\langle A^n v, Q(v,v)\rangle + 2\langle A^n v, F_\perp\rangle. \tag{150}$$

The Killing-precession term has zero trace at every level.

Proof. Differentiate (106) and substitute (146). Since $Q(X,\cdot)$ commutes with A, the viscous contribution is exactly $-2\nu\mathscr{M}^{[s+1]}$. Taking the trace and using Theorem VII.18 gives (150). Finally,

$$\langle A^n v, D(m)v\rangle = 0$$

because $D(m)$ is skew-adjoint and commutes with A^n. □

Corollary VII.20 (Trace-free nonlinear redistribution at the energy level)**.** *At $s = -1$,*

$$\operatorname{tr}\mathscr{C}_{-1}[v; Q(v,v)] = 2\langle v, Q(v,v)\rangle = 0. \tag{151}$$

Also,

$$\operatorname{tr}\mathscr{C}_{-1}[v; D(m)v] = 0. \tag{152}$$

Hence, in the unforced equation, the energy-response tensor obeys

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathscr{M}_v^{[-1]} = -2\nu\mathscr{M}_v^{[0]} + \mathcal{R}_{\mathrm{tf}}(v,m), \qquad \operatorname{tr}\mathcal{R}_{\mathrm{tf}} = 0, \tag{153}$$

where the trace-free term redistributes the six response channels without changing their total high-mode energy.

Proof. The first identity is the standard skew-energy property of incompressible convection. The second follows from skew-adjointness of $D(m)$. □

Remark VII.21. Trace-free does not mean norm-free or dynamically negligible. The nonlinear and Killing terms can substantially change the eigenvectors and eigenvalue distribution of $\mathscr{M}^{[-1]}$ while preserving its trace instantaneously.

I. A response-moment notion of spectral complexity

The Hankel hierarchy gives an exact algebraic notion of the number of active Stokes levels.

Definition VII.22. For a finite-band nonzero field, define its response spectral rank by

$$r_{\mathrm{spec}}(v) := \operatorname{rank}[m_{i+j}]_{i,j\geq 0}. \tag{154}$$

Corollary VII.23. *For finite-band v,*

$$\boxed{r_{\mathrm{spec}}(v) = \#\{k \geq 1 : E_k > 0\}.} \tag{155}$$

This quantity is invariant under $SO(4)$ and under nonzero rescaling of v.

The definition counts distinct Stokes eigenvalues, not the number of signed shells or individual modes. The signed moments h_n provide the additional chirality information required to split each active level.

VIII. CURL-RESPONSE FOURIER TOMOGRAPHY

A. Curl-response moments

Definition VIII.1 (Curl-response tensor)**.** For an integer $n \geq 0$ and sufficiently regular $v \in \mathfrak{K}^\perp$, define the symmetric bilinear form on $\mathfrak{K}$

$$\boxed{\mathscr{C}_n(v)(X,Y) := 2V_R\,\langle Q(X,v), \operatorname{curl}^n Q(Y,v)\rangle\,.} \tag{156}$$

Symmetry follows because curl^n is self-adjoint.

For a single signed shell define its zeroth response weight

$$\mathscr{N}_{k,\sigma}(v) := \mathscr{C}_0(v_{k,\sigma}). \tag{157}$$

It is a positive semidefinite operator on the six-dimensional Killing algebra.

Theorem VIII.2 (Matrix-valued curl moment representation)**.** *Let*

$$v = \sum_{k\geq 1,\sigma=\pm} v_{k,\sigma}$$

and suppose the moment in question is finite. Then

$$\boxed{\mathscr{C}_n(v) = \sum_{k\geq 1,\sigma=\pm} \mu_{k,\sigma}^n \mathscr{N}_{k,\sigma}(v).} \tag{158}$$

Moreover

$$\boxed{\operatorname{tr}\mathscr{N}_{k,\sigma}(v) = \kappa_k \|v_{k,\sigma}\|_2^2} \tag{159}$$

and therefore

$$\boxed{c_n(v) := \operatorname{tr}\mathscr{C}_n(v) = \langle A\operatorname{curl}^n v, v\rangle = \sum_{k,\sigma} \kappa_k \mu_{k,\sigma}^n \|v_{k,\sigma}\|_2^2.} \tag{160}$$

Proof. For every Killing field X,

$$Q(X, v_{k,\sigma}) \in E_{k,\sigma}.$$

Distinct signed shells are orthogonal. Hence the Gram tensor splits over shells, and applying curl^n to the shell response multiplies it by $\mu_{k,\sigma}^n$, proving (158).

Taking $n = 0$ and using (55),

$$\operatorname{tr}\mathscr{N}_{k,\sigma} = 2V_R \sum_{\tau,a} \|T_a^\tau v_{k,\sigma}\|_2^2 = \langle A v_{k,\sigma}, v_{k,\sigma}\rangle,$$

which is (159).

Finally, the same Casimir argument, using commutation with curl, gives

$$2V_R \sum_{\tau,a} \langle T_a^\tau v, \operatorname{curl}^n T_a^\tau v\rangle = \langle A \operatorname{curl}^n v, v\rangle.$$

□

Definition VIII.3 (Response spectral measure)**.** If $\langle Av, v\rangle < \infty$, define the positive $\mathrm{Sym}^+(\mathfrak{K})$-valued measure

$$\boxed{\Sigma_v := \sum_{k\geq 1,\sigma=\pm} \mathscr{N}_{k,\sigma}(v)\,\delta_{\mu_{k,\sigma}}.} \tag{161}$$

Its total trace mass is finite:

$$\operatorname{tr}\Sigma_v(\mathbb{R}) = \langle Av, v\rangle.$$

Equation (158) is precisely the matrix moment formula

$$\mathscr{C}_n = \int_{\mathbb{R}} \lambda^n \, d\Sigma_v(\lambda).$$

B. Block Hankel positivity and response-channel rank

For $N \geq 0$ form the $6(N+1)\times 6(N+1)$ block Hankel matrix

$$\mathbb{H}_N(v) := [\mathscr{C}_{i+j}(v)]_{i,j=0}^N. \tag{162}$$

Theorem VIII.4 (Block Hankel positivity)**.** *Whenever the moments through order $2N$ are finite,*

$$\boxed{\mathbb{H}_N(v) \succeq 0.} \tag{163}$$

Proof. For $X_0, \dots, X_N \in \mathfrak{K}$,

$$\sum_{i,j=0}^N \langle X_i, \mathscr{C}_{i+j} X_j\rangle = \sum_{k,\sigma} \left\langle \sum_i \mu_{k,\sigma}^i X_i, \mathscr{N}_{k,\sigma} \sum_j \mu_{k,\sigma}^j X_j \right\rangle \geq 0.$$

□

The scalar Hankel matrix

$$H_N^{\mathrm{tr}} = [c_{i+j}]_{i,j=0}^N$$

is therefore positive as well.

Theorem VIII.5 (Finite-support rank laws)**.** *Suppose exactly q signed shells $\alpha = 1, \dots, q$ are active, with distinct curl eigenvalues μ_α and nonzero response weights $\mathscr{N}_\alpha$. Then*

$$\boxed{\operatorname{rank} H_N^{\mathrm{tr}} = \min\{N+1, q\}.} \tag{164}$$

For the block Hankel matrix, if $N \geq q-1$,

$$\boxed{\operatorname{rank}\mathbb{H}_N = \sum_{\alpha=1}^q \operatorname{rank}\mathscr{N}_\alpha.} \tag{165}$$

Proof. The scalar weights

$$w_\alpha = \operatorname{tr}\mathscr{N}_\alpha = \kappa_\alpha \|v_\alpha\|_2^2$$

are strictly positive. Thus

$$H_N^{\mathrm{tr}} = V_N \operatorname{diag}(w_1, \dots, w_q) V_N^{\mathsf{T}},$$

where $(V_N)_{i\alpha} = \mu_\alpha^i$. This gives (164).

Similarly,

$$\mathbb{H}_N = \mathbb{V}_N \operatorname{diag}(\mathscr{N}_1, \dots, \mathscr{N}_q) \mathbb{V}_N^*, \tag{166}$$

where $\mathbb{V}_N$ is the block Vandermonde operator with αth block column

$$(1, \mu_\alpha, \dots, \mu_\alpha^N)^{\mathsf{T}} \otimes I_{\mathfrak{K}}.$$

For $N \geq q-1$ the scalar Vandermonde columns are independent, so $\mathbb{V}_N$ is injective on the direct sum of the q copies of $\mathfrak{K}$. Hence multiplication by $\mathbb{V}_N$ and its adjoint preserves the rank of the block diagonal weight. □

Remark VIII.6. The two ranks measure different complexity. The scalar rank counts active signed curl shells. The block rank counts the sum of the independent Killing-response channel ranks carried by those shells.

Corollary VIII.7 (Block-Hankel determinant)**.** *Assume $N = q-1$ and every active $\mathscr{N}_\alpha$ is positive definite on $\mathfrak{K}$. Then*

$$\boxed{\det\mathbb{H}_{q-1} = \left[\prod_{\alpha<\beta}(\mu_\beta - \mu_\alpha)\right]^{12} \prod_{\alpha=1}^q \det\mathscr{N}_\alpha.} \tag{167}$$

Proof. In (166), $\mathbb{V}_{q-1} = V \otimes I_6$ with square Vandermonde V. Therefore

$$\begin{aligned}\det\mathbb{H}_{q-1} &= \det(V\otimes I_6)^2 \prod_\alpha \det\mathscr{N}_\alpha \\ &= \det(V)^{12} \prod_\alpha \det\mathscr{N}_\alpha.\end{aligned}$$

□

C. Finite-band matrix recurrence

Theorem VIII.8 (Annihilating recurrence)**.** *Under the finite-support assumptions of Theorem VIII.5, let*

$$p(z) = \prod_{\alpha=1}^q (z-\mu_\alpha) = z^q + a_{q-1}z^{q-1} + \cdots + a_0. \tag{168}$$

Then the full matrix sequence obeys

$$\boxed{\mathscr{C}_{n+q} + a_{q-1}\mathscr{C}_{n+q-1} + \cdots + a_0\mathscr{C}_n = 0, \qquad n \geq 0.} \tag{169}$$

The same polynomial is the minimal scalar recurrence for $c_n = \operatorname{tr}\mathscr{C}_n$.

Proof. Insert (158). The coefficient of each $\mathscr{N}_\alpha$ is $\mu_\alpha^n p(\mu_\alpha) = 0$. Minimality for the scalar sequence follows because all scalar weights are strictly positive and the nodes are distinct. □

Thus the scalar trace moments identify the signed curl eigenvalues, while the first q matrix moments determine the full response weights through an operator-valued Vandermonde system.

D. The curl-response characteristic function

The spectrum (3) lies in $(1/R)\mathbb{Z}$. Hence

$$e^{i2\pi R\,\mathrm{curl}} = \mathrm{Id} \tag{170}$$

on divergence-free vector fields.

Definition VIII.9 (Operator-valued response characteristic function)**.** For v with finite Stokes energy define

$$\Phi_v(\theta)(X,Y) := 2V_R \left\langle Q(X,v), e^{i\theta\,\mathrm{curl}} Q(Y,v)\right\rangle, \quad \theta \in \mathbb{R}. \tag{171}$$

The function is $2\pi R$-periodic and satisfies

$$\Phi_v(-\theta) = \Phi_v(\theta)^*.$$

Theorem VIII.10 (Exact operator-valued Fourier representation)**.** *For every finite-dissipation v,*

$$\boxed{\Phi_v(\theta) = \sum_{k\geq 1,\sigma=\pm} e^{i\sigma(k+2)\theta/R}\mathscr{N}_{k,\sigma}(v),} \tag{172}$$

where the series converges absolutely in trace norm, hence uniformly in operator norm.

Each shell-response tensor is recovered by Fourier inversion:

$$\boxed{\mathscr{N}_{k,\sigma}(v) = \frac{1}{2\pi R}\int_0^{2\pi R} \Phi_v(\theta) e^{-i\sigma(k+2)\theta/R}\,\mathrm{d}\theta.} \tag{173}$$

Proof. On $E_{k,\sigma}$, $e^{i\theta\,\mathrm{curl}}$ is multiplication by $e^{i\theta\mu_{k,\sigma}}$. Thus (172) follows from shell orthogonality. Moreover

$$\sum_{k,\sigma}\|\mathscr{N}_{k,\sigma}\|_1 = \sum_{k,\sigma}\mathrm{tr}\,\mathscr{N}_{k,\sigma} = \langle Av, v\rangle < \infty.$$

The matrix Fourier series is therefore absolutely convergent in trace norm. Fourier inversion on the circle of length $2\pi R$ gives (173). □

Theorem VIII.11 (Operator-valued positive definiteness)**.** *The periodic function Φ_v is positive definite: for arbitrary $\theta_1,\dots,\theta_m$ and $X_1,\dots,X_m \in \mathfrak{K}$,*

$$\boxed{\sum_{i,j=1}^m \langle X_i, \Phi_v(\theta_i-\theta_j)X_j\rangle_{\mathfrak{K}} \geq 0.} \tag{174}$$

Proof. Use (172):

$$\sum_{i,j}\langle X_i, \Phi(\theta_i-\theta_j)X_j\rangle = \sum_\alpha \left\langle \sum_i e^{-i\mu_\alpha\theta_i}X_i, \mathscr{N}_\alpha \sum_j e^{-i\mu_\alpha\theta_j}X_j\right\rangle \geq 0.$$

□

This is the concrete Navier–Stokes realization of an operator-valued Bochner/Herglotz representation [20].

E. Infinite-band signed-shell energy tomography

Take the trace:

$$\phi_v(\theta) := \mathrm{tr}\,\Phi_v(\theta). \tag{175}$$

By Theorems VIII.2 and VIII.10,

$$\boxed{\phi_v(\theta) = \sum_{k\geq 1,\sigma=\pm}\kappa_k\|v_{k,\sigma}\|_2^2 e^{i\sigma(k+2)\theta/R}.} \tag{176}$$

Theorem VIII.12 (Exact infinite-band energy tomography)**.** *Let v satisfy $\langle Av, v\rangle < \infty$. Then every signed-shell energy is determined by the single scalar periodic function ϕ_v:*

$$\boxed{\|v_{k,\sigma}\|_2^2 = \frac{1}{\kappa_k}\frac{1}{2\pi R}\int_0^{2\pi R}\phi_v(\theta)e^{-i\sigma(k+2)\theta/R}\,\mathrm{d}\theta.} \tag{177}$$

No finite-band hypothesis is required.

Proof. The Fourier coefficient at $n = \sigma(k+2)$ in (176) is $\kappa_k\|v_{k,\sigma}\|_2^2$. Since $k \geq 1$, $\kappa_k > 0$. □

Remark VIII.13 (Forbidden low harmonics). For $v \perp \mathfrak{K}$, the Fourier coefficients of ϕ_v and Φ_v vanish at $n = 0, \pm 1, \pm 2$. The response circle therefore carries a built-in spectral gap between the zero harmonic and the first high-mode frequencies $n = \pm 3$.

Normalize by the total dissipation weight:

$$\psi_v(\theta) := \frac{\phi_v(\theta)}{\phi_v(0)}. \tag{178}$$

Then ψ_v is the characteristic function of the probability distribution

$$p_{k,\sigma} = \frac{\kappa_k\|v_{k,\sigma}\|_2^2}{\langle Av, v\rangle}. \tag{179}$$

Corollary VIII.14 (Dissipation-weighted curl statistics)**.** *If the required moments are finite, then*

$$\frac{1}{i}\psi_v'(0) = \sum_{k,\sigma} p_{k,\sigma}\mu_{k,\sigma}, \tag{180}$$

$$-\psi_v''(0) - |\psi_v'(0)|^2 = \mathrm{Var}_p(\mu) \geq 0. \tag{181}$$

The variance vanishes if and only if the field is supported on one signed curl shell.

F. Stokes dynamics as heat flow on the response circle

Consider

$$\partial_t v + \nu A v = 0. \tag{182}$$

Then

$$v_{k,\sigma}(t) = e^{-\nu\kappa_k t} v_{k,\sigma}(0)$$

and hence

$$\mathscr{N}_{k,\sigma}(t) = e^{-2\nu\kappa_k t}\mathscr{N}_{k,\sigma}(0). \tag{183}$$

Theorem VIII.15 (Matrix heat equation on the response circle)**.** *Under* (182),

$$\boxed{\partial_t \Phi = 2\nu\left(\partial_\theta^2 + \frac{4}{R^2}\right)\Phi} \tag{184}$$

on the circle $\mathbb{R}/(2\pi R\mathbb{Z})$*. The same equation holds for* $\phi = \operatorname{tr}\Phi$*.*

Proof. For one Fourier mode,

$$2\nu\left(\partial_\theta^2 + \frac{4}{R^2}\right)e^{i\mu\theta} = -2\nu\left(\mu^2 - \frac{4}{R^2}\right)e^{i\mu\theta} = -2\nu\kappa e^{i\mu\theta}.$$

Compare with (183) and sum the absolutely convergent Fourier series. □

Remark VIII.16. The auxiliary circle is not physical space. It is the Pontryagin dual of the integer-spaced curl spectrum. The heat equation (184) is therefore an exact spectral representation of viscous relaxation, not an additional spatial dimension.

Corollary VIII.17 (Monotone Fourier coefficient decay)**.** *Every matrix Fourier coefficient evolves independently:*

$$\widehat{\Phi}_{\sigma(k+2)}(t) = e^{-2\nu\kappa_k t}\widehat{\Phi}_{\sigma(k+2)}(0).$$

Thus the response circle diagonalizes the complete matrix-valued Stokes relaxation.

G. What matrix tomography reconstructs

The Fourier inversion theorem recovers $\mathscr{N}_{k,\sigma}$, not necessarily the full vector $v_{k,\sigma}$.

Proposition VIII.18 (High-shell noninjectivity)**.** *Fix* $k \geq 3$ *and* σ*. Consider the normalized quadratic response map*

$$\mathcal{R}_{k,\sigma} : \mathbb{R}P^{d_k-1} \longrightarrow \mathrm{Sym}_6, \qquad [v] \longmapsto \frac{\mathscr{N}_{k,\sigma}(v)}{\|v\|_2^2}. \tag{185}$$

This map cannot be injective.

Proof. The domain is a compact manifold of real dimension

$$d_k - 1 = (k+1)(k+3) - 1.$$

For $k \geq 3$ this is at least 23. The real vector space Sym_6 has dimension 21. A compact manifold of topological dimension greater than 21 cannot embed continuously in $\mathbb{R}^{21}$. Since (185) is continuous, it is not injective. □

Remark VIII.19. Thus even complete knowledge of the response circle recovers the full *response geometry* of every signed shell and its energy, but not the internal phase state of sufficiently high shells. Additional covariants would be required for state-level orbit reconstruction.

H. Information hierarchy, limitations, and relation to prior theory

The combined structure separates four inverse problems.

(i) **State reconstruction.** The response fields form a lossless transform; two fixed same-chirality channels suffice globally.

(ii) **Gram geometry.** The 6×6 response tensor retains dissipation, chiral traces and isometry covariants, but it can lose internal shell phase information.

(iii) **Moment tomography.** The Stokes-weighted hierarchy recovers spectral moments and finite-band energy distributions through Hankel/Vandermonde reconstruction.

(iv) **Fourier tomography.** The curl-response characteristic function recovers every signed-shell response tensor and every signed-shell energy without a finite-band hypothesis.

General g-frames and operator-valued frames were developed independently of fluid mechanics [16, 22]. Matrix-valued moment theory and operator-valued Bochner theorems provide abstract reconstruction frameworks, while the present setting supplies a concrete hydrodynamic measure with support fixed by the round-sphere curl spectrum. The results should therefore be read as an exact geometric realization, not as a claim that frame or moment theory itself is new.

Three limitations are essential. First, Gram compression need not determine a high-dimensional shell state. Second, exact moment inversion from finitely many moments requires finite spectral support or additional determinacy information; the Fourier-circle theorem avoids that issue for finite-dissipation states. Third, none of the global L^2 response identities is an endpoint L^∞ continuation criterion for the three-dimensional Navier–Stokes problem.

IX. OBSERVABLE LEAKAGE, RESPONSE-CIRCLE GENERATORS, AND SPECTRAL SELECTION

The reduction and response theories developed above admit several further consequences once they are used together. This section records those consequences separately because they connect objects that were introduced for different purposes: the Clebsch–Gordan support of nonlinear leakage, minimal Killing observability, the response characteristic function, and the normalized spectral energy distribution.

A. Clebsch–Gordan locality and parity of nonlinear interactions

The complexified signed shell has the representation

$$E^{\mathbb{C}}_{k,\sigma} \simeq V_{j_+} \boxtimes V_{j_-}, \qquad j_\tau = \frac{k+1+\tau\sigma}{2}. \tag{186}$$

Since the Levi-Civita connection and Leray projection are isometry equivariant, Q is an $SO(4)$-equivariant bilinear map. Ref. [2] explicitly left the resulting Clebsch–Gordan shell-selection question open; the next theorem supplies that analysis.

Theorem IX.1 (Interaction parity and spectral locality)**.** *Suppose*

$$\Pi_{\ell,\tau} Q(E_{k_1,\sigma_1}, E_{k_2,\sigma_2}) \neq 0.$$

Then

$$\boxed{\ell \equiv k_1 + k_2 \pmod 2, \qquad \ell \le k_1 + k_2.} \tag{187}$$

More precisely,

$$\boxed{\ell \le k_1 + k_2 + 1 - |\sigma_1 + \sigma_2 - \tau|\,.} \tag{188}$$

Thus, when $\sigma_1 = \sigma_2 = -\tau$, one has the stronger bound

$$\ell \le k_1 + k_2 - 2.$$

Proof. A nonzero projected interaction produces a nonzero $SO(4)$-intertwiner from $E^{\mathbb{C}}_{k_1,\sigma_1} \otimes E^{\mathbb{C}}_{k_2,\sigma_2}$ to $E^{\mathbb{C}}_{\ell,\tau}$. For each $SU(2)$ factor, the Clebsch–Gordan rule therefore requires

$$j_\pm(\ell,\tau) \le j_\pm(k_1,\sigma_1) + j_\pm(k_2,\sigma_2),$$

and the output spin has the same integer/half-integer parity as the sum of the two input spins. Doubling the $+$ spin gives

$$\ell + 1 + \tau \le k_1 + k_2 + 2 + \sigma_1 + \sigma_2,$$

while doubling the $-$ spin gives

$$\ell + 1 - \tau \le k_1 + k_2 + 2 - \sigma_1 - \sigma_2.$$

Taking the smaller upper bound yields (188). Modulo two, $2j_\pm(k,\sigma) \equiv k$, so the Clebsch–Gordan parity condition gives $\ell \equiv k_1 + k_2 \pmod 2$. □

Corollary IX.2 (Finite output band)**.** *If a finite-dimensional candidate space W is supported in shell levels $k \le K$, then*

$$Q(W,W) \subset \bigoplus_{\ell \le 2K,\ \tau=\pm} E_{\ell,\tau}.$$

In particular, every leakage field from W has Stokes eigenvalue at most

$$\kappa_{2K} = \frac{4K(K+2)}{R^2}.$$

B. Arbitrary two-probe observability

The sharp theorem above used an orthonormal pair of Killing probes. The orthogonality assumption is not needed for injectivity.

Theorem IX.3 (Angle-stable two-probe observability)**.** *Let $K, L \in \mathfrak{K}_\tau$ be linearly independent and let*

$$G(K,L) = \begin{pmatrix} \langle K,K\rangle & \langle K,L\rangle \\ \langle L,K\rangle & \langle L,L\rangle \end{pmatrix}$$

be their L^2 Gram matrix. Then every $v \in D(A^{1/2}) \cap \mathfrak{K}^\perp$ satisfies

$$\boxed{2V_R\left(\|Q(K,v)\|_2^2 + \|Q(L,v)\|_2^2\right) \ge \frac{\lambda_{\min}(G(K,L))}{3R^2}\|v\|_2^2.} \tag{189}$$

Hence any two linearly independent same-chirality probes determine the complete non-Killing state.

If K, L are unit probes and θ is their angle, then

$$2V_R\left(\|Q(K,v)\|_2^2 + \|Q(L,v)\|_2^2\right) \ge \frac{1-|\cos\theta|}{3R^2}\|v\|_2^2. \tag{190}$$

Proof. Choose an orthonormal basis e_1, e_2 of $\mathrm{span}\{K,L\}$, and let C be the 2×2 matrix of the coordinates of K, L in that basis. By linearity in the Killing argument,

$$\begin{pmatrix} Q(K,v) \\ Q(L,v) \end{pmatrix} = C \begin{pmatrix} Q(e_1,v) \\ Q(e_2,v) \end{pmatrix}.$$

Therefore

$$\begin{aligned} &\|Q(K,v)\|_2^2 + \|Q(L,v)\|_2^2 \\ &\quad \ge \lambda_{\min}(C^*C)\left(\|Q(e_1,v)\|_2^2 + \|Q(e_2,v)\|_2^2\right). \end{aligned}$$

The nonzero eigenvalues of C^*C are those of $CC^* = G(K,L)$. Apply the sharp orthonormal two-probe bound (66). For unit probes the Gram eigenvalues are $1 \pm |\cos\theta|$. □

There is also a uniform one-probe result on the odd shell sector.

Theorem IX.4 (Uniform one-probe coercivity on odd levels)**.** *Let $K \in \mathfrak{K}_\tau$ be L^2-normalized and let*

$$\mathcal{H}_{\mathrm{odd}} = \widehat{\bigoplus}_{\substack{k\geq 1 \text{ odd} \\ \sigma=\pm}} E_{k,\sigma}.$$

Then, for $v \in D(A^{1/2}) \cap \mathcal{H}_{\mathrm{odd}}$,

$$\boxed{2V_R\|Q(K,v)\|_2^2 \geq \frac{1}{18R^2}\|v\|_2^2.} \tag{191}$$

The constant $1/(18R^2)$ is sharp.

Proof. On $E_{k,\sigma}$, align K with a Cartan generator of the $SU(2)_\tau$ factor. The chiral three-axis response has scalar eigenvalue

$$d_\tau(k,\sigma) = \frac{\kappa_k}{2}\left(1 - \frac{2\tau\sigma}{k+2}\right).$$

For spin $j_\tau = (k+1+\tau\sigma)/2$, a single axis has squared response weights proportional to $m^2/[j_\tau(j_\tau+1)]$. When k is odd, j_τ is half-integer and $|m| \geq 1/2$. Hence the minimum one-axis eigenvalue is

$$\frac{d_\tau(k,\sigma)}{4j_\tau(j_\tau+1)} = \begin{cases} \dfrac{k^2}{2R^2(k+2)^2}, & \tau\sigma = +1, \\ \dfrac{(k+4)^2}{2R^2(k+2)^2}, & \tau\sigma = -1. \end{cases}$$

Over odd $k \geq 1$ and both signs, the minimum is $1/(18R^2)$, attained at $k=1$ and $\tau\sigma = +1$. Shell orthogonality gives (191). □

C. Two-probe detection of the full leakage variety

Let $W \subset \mathfrak{K}^\perp$ be a finite-dimensional Stokes-invariant candidate reduction and define its bilinear leakage map

$$\mathcal{L}_W(u,v) := (I - P_W)Q(u,v), \qquad u,v \in W. \tag{192}$$

For an orthonormal basis $\{e_i\}$ of W, set

$$\Lambda(W) = \sum_{i,j} \|\mathcal{L}_W(e_i,e_j)\|_2^2.$$

The symmetric projected convection has no Killing component: for every Killing field K_0,

$$\langle Q(u,v), K_0\rangle = 0. \tag{193}$$

Indeed, integration by parts followed by the Killing equation cancels the two polarized terms. Thus every leakage field lies in $\mathfrak{K}^\perp$.

Theorem IX.5 (Two-probe leakage certificate)**.** *Let $K, L \in \mathfrak{K}_\tau$ be linearly independent and define*

$$\Lambda_{K,L}(W) := 2V_R \sum_{i,j}\Big(\|Q(K,\mathcal{L}_W(e_i,e_j))\|_2^2 + \|Q(L,\mathcal{L}_W(e_i,e_j))\|_2^2\Big). \tag{194}$$

If W is supported in levels $k \leq K_{\max}$, then

$$\frac{\lambda_{\min}(G(K,L))}{3R^2}\Lambda(W) \leq \Lambda_{K,L}(W) \leq \lambda_{\max}(G(K,L))\kappa_{2K_{\max}}\Lambda(W). \tag{195}$$

Consequently,

$$\boxed{\Lambda(W) = 0 \iff \Lambda_{K,L}(W) = 0.} \tag{196}$$

Moreover, if Φ is the algebraic leakage section on a product of Grassmannians and

$$\Psi_{K,L} := \big(Q(K,\cdot), Q(L,\cdot)\big) \circ \Phi,$$

then at every zero-leakage point

$$\boxed{\ker d\Psi_{K,L} = \ker d\Phi, \qquad \operatorname{rank} d\Psi_{K,L} = \operatorname{rank} d\Phi.} \tag{197}$$

Thus two Killing-response channels detect not only exact closure but also the full infinitesimal deformation geometry of the leakage variety.

Proof. Apply Theorem IX.3 to each leakage field and sum over i,j. This gives the lower bound. For the upper bound, choose an orthonormal basis of $\operatorname{span}\{K,L\}$. The largest eigenvalue of $G(K,L)$ bounds the pair response by that orthonormal pair, and the latter is bounded by the full six-channel Casimir sum. Hence

$$\Lambda_{K,L}(W) \leq \lambda_{\max}(G(K,L)) \sum_{i,j}\big\langle A\mathcal{L}_W(e_i,e_j), \mathcal{L}_W(e_i,e_j)\big\rangle.$$

By Theorems IX.1 and IX.2, every leakage field is supported in levels at most $2K_{\max}$, yielding the upper estimate.

The response pair is injective on $\mathfrak{K}^\perp$, so composing the leakage section with it does not change the zero set. At a zero, $d\Psi_{K,L}$ is the fixed injective response pair composed with $d\Phi$. An injective postcomposition preserves both kernel and rank. □

Corollary IX.6 (One-probe closure test for opposite shell parity)**.** *Let*

$$W = L_\alpha \oplus L_\beta, \qquad L_\alpha \subset E_{k_1,\sigma_1}, \quad L_\beta \subset E_{k_2,\sigma_2},$$

with k_1+k_2 odd. For any L^2-normalized nonzero $K \in \mathfrak{K}_\tau$, define

$$\Lambda_K(W) = 2V_R\|Q(K,\mathcal{L}_W)\|_{\mathrm{HS}}^2.$$

Then

$$\boxed{\frac{1}{18R^2}\Lambda(W) \leq \Lambda_K(W) \leq \kappa_{k_1+k_2}\Lambda(W),} \tag{198}$$

and hence

$$W \text{ is exact} \iff \Lambda_K(W) = 0.$$

Proof. Single-shell terms vanish identically, so only the cross interaction $Q(L_\alpha, L_\beta)$ remains. By Theorem IX.1, every output shell has odd index because k_1+k_2 is odd, and its level is at most k_1+k_2. Apply Theorem IX.4 and the Casimir upper bound. □

D. The response circle as a master generating object

Let

$$\mathscr{D}_\theta = -i\partial_\theta, \qquad \mathscr{L}_\theta = \mathscr{D}_\theta^2 - \frac{4}{R^2} = -\partial_\theta^2 - \frac{4}{R^2}$$

on the circle of length $2\pi R$, restricted to the high harmonics $|n| \geq 3$. On the Fourier character $e^{i\mu_{k,\sigma}\theta}$,

$$\mathscr{D}_\theta = \mu_{k,\sigma}, \qquad \mathscr{L}_\theta = \kappa_k.$$

Theorem IX.7 (Master response-circle functional calculus)**.** *Let Φ_v be the operator-valued response characteristic function from Theorem VIII.10. Whenever the corresponding moments are finite,*

$$\boxed{\mathscr{C}_n(v) = \mathscr{D}_\theta^n \Phi_v(\theta)|_{\theta=0}\,,} \tag{199}$$

$$\boxed{\mathscr{M}_v^{[s]} = \mathscr{L}_\theta^s \Phi_v(\theta)|_{\theta=0}\,.} \tag{200}$$

For noninteger s, $\mathscr{L}_\theta^s$ is the positive Fourier multiplier on the high-harmonic subspace.

Define the energy-response characteristic function

$$\mathscr{E}_v(\theta) := \mathscr{L}_\theta^{-1}\Phi_v(\theta), \qquad \varepsilon_v(\theta) := \operatorname{tr}\mathscr{E}_v(\theta). \tag{201}$$

Then

$$\boxed{\varepsilon_v(\theta) = \sum_{k\geq 1,\sigma=\pm} \|v_{k,\sigma}\|_2^2 e^{i\sigma(k+2)\theta/R}.} \tag{202}$$

Consequently,

$$\|v\|_2^2 = \varepsilon_v(0), \tag{203}$$

$$\langle \operatorname{curl}^r v, v\rangle = \mathscr{D}_\theta^r \varepsilon_v|_{\theta=0}\,, \tag{204}$$

$$\langle A^n v, v\rangle = \mathscr{L}_\theta^n \varepsilon_v|_{\theta=0}\,, \tag{205}$$

$$\langle A^n \operatorname{curl}^{-1} v, v\rangle = \mathscr{D}_\theta^{-1}\mathscr{L}_\theta^n \varepsilon_v\big|_{\theta=0}\,. \tag{206}$$

Thus the Stokes hierarchy, curl hierarchy, kinetic energy, helicity and signed inverse-curl moments are all jets or Fourier multipliers of one periodic function.

Proof. The Fourier representation

$$\Phi_v(\theta) = \sum_{k,\sigma} e^{i\mu_{k,\sigma}\theta}\mathscr{N}_{k,\sigma}$$

and the identities

$$\mathscr{D}_\theta e^{i\mu\theta} = \mu e^{i\mu\theta}, \qquad \mathscr{L}_\theta e^{i\mu\theta} = \left(\mu^2 - \frac{4}{R^2}\right)e^{i\mu\theta} = \kappa e^{i\mu\theta}$$

give (199)–(200). Since $\operatorname{tr}\mathscr{N}_{k,\sigma} = \kappa_k\|v_{k,\sigma}\|_2^2$, applying $\mathscr{L}_\theta^{-1}$ and taking the trace gives (202). The remaining identities are its Fourier moments. □

E. Extreme spectral characters and exact same-shell sectors

For $v \neq 0$, normalize

$$\chi_v(\theta) := \frac{\varepsilon_v(\theta)}{\varepsilon_v(0)} = \sum_{k,\sigma} p_{k,\sigma} e^{i\sigma(k+2)\theta/R}, \tag{207}$$

$$p_{k,\sigma} := \frac{\|v_{k,\sigma}\|_2^2}{\|v\|_2^2}. \tag{208}$$

This is a positive-definite function and the characteristic function of a probability measure on the allowed signed curl lattice.

Theorem IX.8 (Spectral extremality of same-shell exact states)**.** *Within the convex set of normalized energy-response functions (207), χ_v is an extreme point if and only if v is supported on one signed curl shell. In that case*

$$\boxed{\chi_v(\theta) = e^{i\sigma(k+2)\theta/R},} \tag{209}$$

and $Q(v,v) = 0$.

Thus every Grassmannian family $\operatorname{Gr}(d, E_{k,\sigma})$ of exact same-shell reductions is collapsed by scalar spectral tomography to the single extreme character (209).

Proof. The Fourier coefficients in (207) form a probability vector. The extreme points of a probability simplex are the Dirac masses, hence the extreme positive-definite functions are the single characters. A Dirac spectral measure is equivalent to support in one signed shell. The single-shell cancellation then gives $Q(v,v) = 0$. □

F. Fisher–Rao Stokes selection

The normalized shell energies reveal a second geometry. General Fisher–Rao gradient-flow theory is independent of the present fluid construction [8]; here it arises exactly from Stokes spectral decay.

Theorem IX.9 (Fisher–Rao spectral selection law)**.** *Consider the linear Stokes evolution*

$$\partial_t v + \nu A v = 0$$

and set

$$E_\alpha(t) = \|v_\alpha(t)\|_2^2,$$

$$p_\alpha(t) = \frac{E_\alpha(t)}{\sum_\beta E_\beta(t)},$$

$$\bar\kappa(t) = \sum_\alpha p_\alpha(t)\kappa_\alpha.$$

where $\alpha = (k,\sigma)$. Then

$$\boxed{\dot p_\alpha = -2\nu\big(\kappa_\alpha - \bar\kappa\big)p_\alpha.} \tag{210}$$

On every finite spectral simplex, if

$$g_p(\xi,\eta)=\sum_\alpha \frac{\xi_\alpha\eta_\alpha}{p_\alpha}$$

is the Fisher–Rao metric and $\mathcal{K}(p)=\sum_\alpha \kappa_\alpha p_\alpha$*, then*

$$\boxed{\dot p=-2\nu\ \mathrm{grad}_{FR}\mathcal{K}.} \tag{211}$$

Moreover,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\bar\kappa=-2\nu\,\mathrm{Var}_p(\kappa)\le 0.} \tag{212}$$

Equality holds exactly when the normalized energy is supported on one Stokes level.

Proof. The shell energies satisfy

$$E_\alpha(t)=E_\alpha(0)e^{-2\nu\kappa_\alpha t}.$$

Differentiating their normalization gives (210). For a tangent vector ξ with $\sum_\alpha \xi_\alpha=0$,

$$d\mathcal{K}[\xi]=\sum_\alpha \kappa_\alpha\xi_\alpha=g_p\big(\big(p_\alpha(\kappa_\alpha-\bar\kappa)\big)_\alpha,\xi\big)\,,$$

which proves (211). Finally,

$$\dot{\bar\kappa}=-2\nu\sum_\alpha p_\alpha(\kappa_\alpha-\bar\kappa)\kappa_\alpha=-2\nu\,\mathrm{Var}_p(\kappa).$$

□

Corollary IX.10 (Exact viscous spectral selection)**.** *Let* k_* *be the smallest active Stokes level and let*

$$P_*(0)=p_{k_*,+}(0)+p_{k_*,-}(0)>0.$$

If $\Delta\kappa$ *is the gap from* κ_{k_*} *to the next active Stokes eigenvalue, then*

$$1-P_*(t)\le\frac{1-P_*(0)}{P_*(0)}e^{-2\nu\Delta\kappa t}. \tag{213}$$

The ratio $p_{k_*,+}(t):p_{k_*,-}(t)$ *is constant. Thus viscosity selects the lowest active Stokes* level*, but does not by itself equalize or select its two chiralities.*

G. Nonlinear response-circle transfer

The same normalized spectral picture survives the full nonlinear equation as an exact decomposition into a viscous gradient term and an energy-conserving transfer term.

For a sufficiently regular unforced high-mode solution

$$\partial_t v+\nu Av+Q(v,v)+D(m)v=0, \tag{214}$$

define

$$\mathcal{T}_{k,\sigma}:=-2\,\langle\Pi_{k,\sigma}Q(v,v),v_{k,\sigma}\rangle \tag{215}$$

and its response-circle transform

$$\mathcal{T}_v(\theta):=\sum_{k,\sigma}\mathcal{T}_{k,\sigma}e^{i\sigma(k+2)\theta/R}. \tag{216}$$

Theorem IX.11 (Nonlinear response-circle balance)**.** *The energy-response characteristic function* ε_v *satisfies*

$$\boxed{\partial_t\varepsilon_v=2\nu\left(\partial_\theta^2+\frac{4}{R^2}\right)\varepsilon_v+\mathcal{T}_v.} \tag{217}$$

The nonlinear transfer source obeys the two exact conservation constraints

$$\boxed{\mathcal{T}_v(0)=0,\qquad \mathscr{D}_\theta\mathcal{T}_v|_{\theta=0}=0.} \tag{218}$$

They are respectively the energy and helicity conservation laws of the Euler nonlinearity.

If $E(t)=\|v(t)\|_2^2$ *and* $p_\alpha=E_\alpha/E$*, then*

$$\boxed{\dot p_\alpha=-2\nu(\kappa_\alpha-\bar\kappa)p_\alpha+\frac{\mathcal{T}_\alpha}{E}.} \tag{219}$$

Thus normalized spectral dynamics decomposes exactly into a Fisher–Rao viscous gradient flow and a zero-sum nonlinear redistribution. In particular,

$$\boxed{\dot{\bar\kappa}=-2\nu\,\mathrm{Var}_p(\kappa)+\frac{1}{E}\sum_\alpha\kappa_\alpha\mathcal{T}_\alpha.} \tag{220}$$

Proof. The Killing drift preserves each signed shell and is skew-adjoint there, so it contributes nothing to the shell energies. Taking the L^2 inner product of (214) with $v_{k,\sigma}$ gives

$$\dot E_{k,\sigma}=-2\nu\kappa_kE_{k,\sigma}+\mathcal{T}_{k,\sigma}.$$

Fourier summation and

$$2\nu\left(\partial_\theta^2+\frac{4}{R^2}\right)e^{i\mu_{k,\sigma}\theta}=-2\nu\kappa_ke^{i\mu_{k,\sigma}\theta}$$

give (217).

The zeroth transfer moment is

$$\sum_{k,\sigma}\mathcal{T}_{k,\sigma}=-2\langle Q(v,v),v\rangle=0.$$

The first signed-curl moment is

$$\sum_{k,\sigma}\mu_{k,\sigma}\mathcal{T}_{k,\sigma}=-2\langle Q(v,v),\mathrm{curl}\,v\rangle=0,$$

because $\nabla_v v=\mathrm{grad}_g(|v|^2/2)-v\times\mathrm{curl}\,v$: the gradient is orthogonal to $\mathrm{curl}\,v$ and $(v\times\mathrm{curl}\,v)\cdot\mathrm{curl}\,v=0$. This proves (218).

Normalize the shell-energy equation by $E(t)$. Since the nonlinear transfer sums to zero, $\dot E=-2\nu\bar\kappa E$, which gives (219). Multiplication by κ_α and summation gives (220). □

Corollary IX.12 (Finite-band conservation factorization)**.** *If* v *has finite spectral support, write* $z=e^{i\theta/R}$*. Then the Laurent polynomial*

$$P_v(z)=\sum_{k,\sigma}\mathcal{T}_{k,\sigma}z^{\sigma(k+2)}$$

has a double root at $z = 1$. Hence

$$\boxed{P_v(z) = (z-1)^2 S_v(z)} \tag{221}$$

for another Laurent polynomial S_v.

H. Quantitative information loss under Gram compression

The two-probe transform is injective, whereas the pure-shell Gram tensor is necessarily many-to-one at high shell number. The chiral trace constraints sharpen the dimension obstruction.

Theorem IX.13 (Lower bound on Gram-compression fiber dimension)**.** *Fix $k \geq 1$ and σ, and consider the normalized pure-shell Gram map*

$$\Gamma_{k,\sigma} : \mathbb{R}P^{d_k-1} \longrightarrow \mathrm{Sym}_6, \qquad [v] \longmapsto \frac{\mathscr{M}_v}{\|v\|_2^2}.$$

Its image lies in the affine subspace

$$\begin{aligned}\mathcal{A}_{k,\sigma} = \big\{ M \in \mathrm{Sym}_6 : \ & \mathrm{tr}_{\mathfrak{K}_+} M = d_+^{(k,\sigma)}, \\ & \mathrm{tr}_{\mathfrak{K}_-} M = d_-^{(k,\sigma)} \big\}.\end{aligned} \tag{222}$$

which has dimension 19.

On every constant-rank stratum of $\Gamma_{k,\sigma}$, every local fiber has dimension at least

$$\boxed{d_k - 20 = k^2 + 4k - 17.} \tag{223}$$

In particular, for $k \geq 3$ Gram compression loses at least four continuous projective degrees of freedom at every regular point of maximal differential rank.

Proof. The pure-shell chiral partition fixes the two block traces independently after normalization by $\|v\|_2^2$. Since $\dim \mathrm{Sym}_6 = 21$, the two independent affine trace constraints reduce the ambient dimension to 19.

The domain has dimension $d_k - 1$. On a constant-rank stratum with differential rank $r \leq 19$, the constant-rank theorem gives local fiber dimension

$$d_k - 1 - r \geq d_k - 20.$$

For $k = 3$, this lower bound is $24 - 20 = 4$, and it increases quadratically with k. □

Corollary IX.14 (Exact information ladder on a pure shell)**.** *For a fixed signed shell $E_{k,\sigma}$:*

(i) scalar normalized spectral tomography is constant on $\mathbb{R}P^{d_k-1}$;

(ii) normalized Gram compression has local fibers of dimension at least $d_k - 20$ on every constant-rank stratum;

(iii) any two linearly independent same-chirality raw Killing responses are injective.

Thus information loss occurs at compression, not at the Killing-response analysis itself.

X. OPEN PROBLEMS

The unified structure isolates several concrete problems.

(1) Determine the principal orbit types and singular strata of $\mathrm{Gr}(d, E_{k,\sigma})/SO(4)$, and compute the first nontrivial irreducible components of the multi-shell zero-leakage variety.

(2) Classify when the constant-drift Grassmannian geodesic condition is compatible with nontrivial resonance-free shell splitting.

(3) Determine exact frame bounds for arbitrary nonorthogonal pairs of Killing probes and the minimal spatial sampling needed to reconstruct their response fields.

(4) Classify the image and orbit strata of the pure-shell Gram map $v \mapsto \mathscr{M}_v/\|v\|_2^2$, including extremal spectra and cross-chiral ranks.

(5) Develop stable noisy versions of the finite-band Hankel reconstruction and the infinite-band response-circle Fourier inversion.

(6) Use the nonlinear tensor balance and leakage geometry together to determine when cross-shell interaction creates new response spectral rank.

(7) Find higher $SO(4)$-covariants that complement the Gram tensor and separate generic isometry orbits inside high signed-curl shells.

(8) Study persistence of the reduction moduli, response frame, and response circle under perturbations of the round metric.

XI. CONCLUSION

The Killing–Beltrami structure on the round three-sphere organizes both the exact subsystems of Navier–Stokes flow and the information contained in their symmetry responses.

On the reduction side, every subspace of a signed curl shell is an exact pressure-linear Navier–Stokes system. These systems form Grassmannian moduli, acquire exact projector dynamics under Killing drift, and extend to multi-shell reductions as the zero set of a real-algebraic leakage functional. Constant drift admits an exact geodesic criterion, an explicit shell spectrum, finite resonance arithmetic, and special-orthogonal transport.

On the reconstruction side, the six Killing responses factor the Stokes operator and, after normalization, form a lossless Parseval transform. Two same-chirality probes already determine the complete non-Killing velocity, while the one-probe problem has an exact shell-parity law. Gram compression produces a finite-dimensional response geometry; Stokes weighting produces a positive

moment hierarchy; curl weighting produces a positive matrix-valued spectral measure whose periodic characteristic function admits exact Fourier inversion.

The common chain is therefore

$$\begin{aligned}\text{signed-shell structure} &\longrightarrow \text{exact reductions}\\ &\longrightarrow \text{Killing transport}\\ &\longrightarrow \text{response tomography.}\end{aligned}$$

This is one geometric-spectral theory rather than two independent manuscripts.

AI-ASSISTED RESEARCH METHOD

OpenAI ChatGPT (GPT-5.6 Sol) was used as an AI-assisted research tool to generate candidate algebraic deductions, test derivation paths, perform algebraic consistency checks, and organize targeted literature searches. The tool was directed through explicit mathematical prompts specifying the relevant hypotheses, operators, and target identities. Candidate statements were retained only when supported by explicit derivations in the manuscript or by the cited sources; unsupported candidate claims were discarded. The authors independently reviewed the retained AI-assisted mathematical claims and source attributions and assume full responsibility for their accuracy and interpretation.

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In addition to the research-method use disclosed above, OpenAI ChatGPT (GPT-5.6 Sol) assisted with manuscript organization and LaTeX preparation. The authors remain fully responsible for the accuracy, interpretation, and scientific content of the manuscript.

DATA AVAILABILITY

No data were created or analyzed in this study.

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